

Active Geometric Maintenance of the Spinal S-Curve Against Gravity: An Information–Mechanical Coupling Model for Adolescent Idiopathic Scoliosis Onset

Dr. Sayuj Krishnan S, MBBS, DNB (Neurosurgery)
Consultant Neurosurgeon and Spine Surgeon, Yashoda Hospitals, Malakpet, Hyderabad, India
dr.sayujkrishnan@gmail.com

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Abstract

Purpose: Adolescent idiopathic scoliosis (AIS) clusters at peak height velocity, in the thoracic spine, and overwhelmingly in girls (8:1), yet no single framework unifies its timing, location, and sex predominance. We test whether AIS can be modelled as failure of an active, gravity-loaded postural-control system.

Methods: We model the developing spine as a biomechanical control system: a Cosserat rod whose segment-wise stiffness is set by developmental HOX-domain patterning and stabilised against gravity by proprioceptive feedback. Four research strands: (i) cross-species allometric analysis across 12 vertebrates (mouse to dolphin) using $\mathcal{B}_g = EI/MgL^2$; (ii) AlphaFold-derived structural metrics (elongation, hinge-density per 100 residues; length-matched controls) for 61 candidate proteins; (iii) a delay-differential postural-control model through the growth spurt; (iv) synthetic-cohort calibration against published Lonstein & Carlson and BrAIST trial rates (not patient-level records), plus an open SpineWeb/AASCE landmark-geometry consistency check ($n = 534$ usable AP cases).

Results: Cross-species, only humans operate at $\mathcal{B}_g \approx 0.01$, where passive bending stiffness is insufficient and active proprioceptive maintenance is mandatory. A curated subset of extended-filament/channel mechanosensors (Vimentin, Lamin A/C, PIEZO2, SCRIB, PTK7) is 72% more structurally elongated than metabolic regulators ($p = 0.011$, Cohen's $d = 1.19$, $n = 23$; signal dilutes with broader gene sets) and has 70% fewer hinge regions per residue ($p = 0.003$, $n = 61$) — consistent with cooperative load-sensing. In the postural-control simulation, proprioceptive delay scales with body length while neural conduction velocity does not; the resulting feedback-delay instability closes the simulated control margin at the age window of AIS onset, and the model-derived energy deficit correlates strongly with simulated Cobb-angle severity across the growth window ($r = 0.983$, $p = 2.7 \times 10^{-22}$). The effective Bio-Gravitational parameter $\mathcal{B}_g^{\text{eff}}$ correlates strongly with the Lonstein & Carlson Progression Factor ($r = -0.483$, $p < 10^{-50}$). Simulated bracing (modeled as 37% gravitational load-sharing) replicates the overall success rates of the BrAIST trial (72% braced vs 48% observation success) and matches subgroup trends. A linear regression of simulated tissue anisotropy against the critical spine length L_{crit} shows protective scaling ($R^2 = 0.775$, $p < 10^{-17}$), saturating at anisotropy ≥ 6 .

Conclusions: The framework reframes AIS as a predictable failure of an active biomechanical control system, integrating developmental patterning, proprioceptive feedback, and tissue mechanics. The Bio-Gravitational parameter is a candidate biomechanical proxy for published progression/bracing rates; patient-level AIS records are not analysed here. Three hypothesis-generating predictions follow, presented *not* as clinical recommendations: (i) proprioceptive latency should rise non-linearly through the growth spurt in at-risk adolescents; (ii) the model predicts that mitochondrial NAD^+ availability modulates simulated curve progression — **this does not endorse nutraceutical use outside an IRB-approved**

trial; (iii) ciliary length and cellular dithering should increase in scoliotic osteoblasts before macroscopic onset.

Impact Statement: We identify a quantitative scaling law governing the metabolic cost of actively maintaining spinal geometry against gravity, explaining why the human spine is uniquely vulnerable to scoliosis during adolescent growth and reframing AIS as a predictable failure of an active gravity-loaded postural-control system rather than a stochastic genetic defect.

Model Summary: The Allometric Trap

The Question: What physical law governs how organisms maintain shape against gravity, and why does this maintenance fail during adolescent growth?

The Scaling Law: The Bio-Gravitational Number $\mathcal{B}_g = EI/(MgL^2)$ predicts that humans ($\mathcal{B}_g \approx 0.01$) cannot passively resist gravity and must actively maintain spinal geometry at metabolic cost.

The Mechanism:

- **Metabolic Scaling Mismatch:** The cost of anti-gravitational form maintenance scales as L^4 , while biological supply scales as L^2 , creating a critical Energy Deficit Window at $L > 0.35$ m.
- **Molecular Asymmetry (Exploratory):** Initial AlphaFold analysis suggests mechanosensory proteins may exhibit higher structural anisotropy than their metabolic regulators. We present this as a hypothesis-generating observation for future study, acknowledging current AI limitations in predicting mechanical properties.
- **Scoliosis as Scaling Failure:** During adolescent growth, this deficit forces collapse from the active S-curve into a lower-energy scoliotic configuration — a predictable failure of the active control system, not a stochastic genetic defect.

1 Introduction

1.1 The Clinical Problem: Adolescent Idiopathic Scoliosis

Adolescent Idiopathic Scoliosis (AIS) affects 2–4% of adolescents worldwide, representing the most common spinal deformity requiring clinical intervention [1, 2]. Despite over a century of research, the fundamental question remains unanswered: *why does scoliosis initiate during the growth spurt at ages 11–15, and why is the thoracic spine (T8–T10) most vulnerable?*

Current explanatory frameworks — neuromuscular asymmetry [3], genetic predisposition [4], and Hueter-Volkman asymmetric loading [5] — describe *correlations* but offer no unified mechanism connecting age window, anatomic localization, and 8:1 female predominance. As a result, treatment remains empirical: observation for mild curves (Cobb angle $< 25^\circ$), bracing for moderate curves (25 – 45°), and corrective fusion surgery for severe deformity ($> 45^\circ$) [6].

1.2 An Allometric Vulnerability: Why the Human Spine Is Uniquely Exposed

We propose a paradigm shift: AIS is not a random genetic error but a *predictable scaling failure* of an active gravity-loaded control system. The human spine sustains an intricate S-shaped

profile (cervical/lumbar lordosis, thoracic kyphosis) rather than passively sagging into the C-curve predicted by classical beam mechanics [7, 8]. Maintaining this upright posture against gravity is intrinsically metabolically expensive, requiring continuous mechanosensory feedback via proteins like PIEZO1/2 and primary cilia [9–11].

Maintaining an upright posture in a gravitational field is intrinsically expensive, requiring continuous muscular actuation, proprioceptive feedback, and cellular mechanotransduction to oppose gravity and prevent structural collapse [12–15]. Mechanosensory structures, including primary cilia and highly anisotropic demand-side proteins like PIEZO1/2 and Vimentin, act as critical load sensors, continuously monitoring mechanical strain and mediating metabolic responses [9, 10, 16]. Disruptions in this mechanosensory loop, whether through microgravity unloading or genetic mutations in ciliary or mechanotransductive pathways, lead to profound structural abnormalities such as scoliosis and bone density loss [11, 17, 18]. Consequently, spinal morphogenesis and geometry maintenance cannot be viewed solely as a solid mechanics problem, but as a continuous mechanobiological computation.

1.3 The Bridge: Counter-Curvature as an Active Control System

To understand how the spine maintains its shape and why it is uniquely vulnerable during growth, we model the spine as an active control system that continuously expends metabolic energy to oppose gravity. We refer to this active maintenance as “biological counter-curvature” to distinguish it from passive structural support. Like other dissipative biological control systems [19, 20], the spine sustains its geometry only as long as the metabolic budget supports the proprioceptive feedback loop.

This active control system provides a physical basis for its developmental vulnerability. We formalize this through the **Bio-Gravitational Number**, $\mathcal{B}_g = EI/(MgL^2)$, a dimensionless ratio analogous to the Froude number. Cross-species analysis reveals that most vertebrates maintain $\mathcal{B}_g > 0.1$ (passively stable), but humans occupy a unique “Allometric Trap” ($\mathcal{B}_g \approx 0.01$) where passive stiffness is entirely insufficient and active metabolic maintenance is required [21, 22]. This fragile state requires constant mechanosensory monitoring.

During the adolescent growth spurt, this control system faces a fundamental scaling crisis. Established models of human postural control demonstrate that the upright state is stabilized by proportional-derivative (PD) feedback, subject to a time delay τ [23, 24]. As the spine elongates rapidly, the absolute neural feedback delay τ increases because nerve conduction velocity does not scale proportionately [25].

When this neural delay exceeds a critical threshold, the system undergoes a supercritical Hopf bifurcation [26]. We term this the **Derivative Gain Trap**, where derivative feedback that provides essential damping during childhood becomes destabilizing, triggering continuous, micro-mechanical scoliotic buckling events.

Furthermore, this delay instability is exacerbated by an **Energy Deficit Window**. The metabolic cost of maintaining the active control against gravity scales as L^4 , while nutrient supply to the avascular intervertebral disc scales only as L^2 [27, 28]. We propose that this transient energy deficit, potentially driving localized NAD⁺ insufficiency and axonal degradation [29, 30], further increases proprioceptive delay and precipitates Adolescent Idiopathic Scoliosis (AIS). The intersection of increased neural delay and restricted metabolic capacity during peak height velocity (PHV) forms the biophysical foundation of scoliosis onset.

1.4 Predictions

By modeling the spine through a Delay Differential Equation (DDE) control framework, this model moves beyond correlative observations to a candidate causal mechanism, generating three specific, falsifiable theoretical predictions for hypothesis-generation. We emphasize that these

predictions arise from in-silico modeling and **do not constitute clinical recommendations**; prospective testing in IRB-approved cohorts is required before any intervention is suggested:

- **Theoretical metabolic rescue:** The model predicts that mitochondrial cofactor availability (e.g., NAD^+) modulates curve progression in simulation. This generates a hypothesis testable in IRB-approved adolescent cohorts; we do not endorse off-label supplementation in adolescents outside such trials.
- **Neuromotor delay as a biomarker:** Longitudinal measurements of spinal proprioceptive latency τ are predicted to rise non-linearly during the adolescent growth spurt in adolescents who later progress to macroscopic curves, providing a candidate pre-clinical biomarker testable with standard somatosensory evoked potential studies.
- **Ciliary length and cellular dithering:** If structural sensing requires active metabolic supply, the model predicts cells under metabolic constraint will upregulate ciliary length or rely on stochastic micro-tremors to maintain signal-to-noise ratios. These are testable in scoliotic osteoblast samples obtained from existing intraoperative tissue banks.

2 Theoretical Framework

2.1 The Bio-Gravitational Number: A Stability Threshold

The central question is: under what physical conditions can an organism maintain its shape against gravity? We derive a dimensionless parameter, the **Bio-Gravitational Number**:

$$\mathcal{B}_g = \frac{EI}{MgL^2}, \quad (1)$$

where EI is spinal flexural stiffness, M is body mass, g is gravitational acceleration, and L is spine length. This ratio compares passive structural resistance to gravitational bending load.

Cross-species analysis reveals a critical threshold at $\mathcal{B}_g \approx 0.1$. Species above this value (most quadrupeds) can resist gravity passively through bone and disc stiffness alone. Humans occupy a unique position: adult $\mathcal{B}_g \approx 0.01$, placing us deep within the “Allometric Trap” where passive resistance fails and the normal spinal S-curve (cervical lordosis, thoracic kyphosis, lumbar lordosis) requires continuous metabolic maintenance via cellular strain sensors (PIEZO1/2, primary cilia) and proprioceptive feedback.

2.2 The Energy Deficit Window

During adolescent growth, two unfavorable scaling relationships collide. The metabolic cost of maintaining the S-curve against gravity scales approximately as L^4 (bending moment grows as L^2 , curvature correction effort scales as L^2). Nutrient supply to avascular intervertebral discs, governed by diffusion and convective transport, scales more slowly as L^2 (surface area limited) [27, 28].

These curves cross at a critical spine length $L_{\text{crit}} \approx 0.35$ m. For $L > L_{\text{crit}}$, the system enters an **Energy Deficit Window** where metabolic demand exceeds supply capacity. At $L = 0.45$ m (typical 15-year-old), the deficit reaches approximately 41%. This age-length correspondence ($L_{\text{crit}} \approx 0.35$ m $\rightarrow$ ages 11–12) matches the well-known clinical peak incidence of Adolescent Idiopathic Scoliosis without parameter fitting to clinical data.

2.3 Information-Elasticity Coupling: HOX Genes to Geometry

The model formalizes how developmental patterning genes (particularly HOX genes encoding vertebral identity along the cranio-caudal axis) translate into mechanical geometry. We introduce an **Information Field** $I(s)$ representing the spatial concentration of HOX-encoded

positional information along the spine (coordinate s). This field couples to spine mechanics through three channels (see Supplementary Theory for full derivation):

1. **Rest curvature modulation:** Information gradients specify target curvatures (lordotic at cervical/lumbar, kyphotic at thoracic).
2. **Stiffness modulation:** Information modulates local bending and torsional stiffness via extracellular matrix composition.
3. **Active moment generation:** Information gradients drive active muscular moments maintaining the S-curve.

When the energy deficit exceeds a critical threshold, this active control system can no longer suppress small left-right asymmetries (inevitable developmental noise, typically $\sim 1\text{--}5\%$). These asymmetries, benign during childhood, are amplified into macroscopic scoliotic deformities (Cobb angles $> 15^\circ$) during the deficit window.

2.4 Why Lateral (Coronal Plane) Buckling?

The spine is loaded primarily in the sagittal plane (forward-backward) by gravity. Why does deformity emerge laterally? The model predicts a **directional-stiffness anisotropy**: developmental patterning gradients (encoding the S-curve) are aligned with gravitational stress in the sagittal plane, providing maximal effective stiffness in that direction. Small lateral asymmetries create orthogonality between information and stress vectors, reducing effective lateral stiffness by up to 80%. This creates a localized “soft spot” at the thoracolumbar junction (T8–T10, corresponding to steepest information gradient), directing instability into the coronal plane and explaining the anatomic localization of right thoracic AIS.

Full mathematical derivations, stability analysis, and parameter sensitivity studies are provided in Supplementary Theory Sections S1–S4.

3 Methods

3.1 Computational Framework

We implemented the Information-Elasticity Coupling framework using two complementary approaches: (1) a fast deterministic beam model for parameter space exploration, and (2) full 3D Cosserat-rod simulations (a standard slender-continuum model in spine biomechanics) capturing both bending and twisting under large deformations.

For 3D simulations, we used **PyElastica** [31, 32], an open-source Python implementation of Cosserat rod theory. The spine was modeled as a slender elastic rod ($n = 50\text{--}100$ elements) with length $L = 0.3\text{--}0.5$ m, radius $r = 0.01$ m, Young’s modulus $E_0 = 1$ GPa, and density $\rho = 1100$ kg/m³ (representative of vertebra-disc composite properties). Boundary conditions: clamped at base (sacrum), free at top (cranium), under uniform gravitational loading $\mathbf{g} = -9.81$ m/s².

As an optional acceleration layer, we implemented a GPU-native NVIDIA Warp screening workflow for large Euler–Bernoulli parameter sweeps. This workflow evaluates analytic passive-sag and $\mathcal{B}_g$ quantities across candidate length, radius, stiffness, and density combinations, then forwards selected high-risk regimes to the validated PyElastica beam setup for nonlinear refinement. Warp results are used only for computational triage and reproducibility auditing; all mechanistic claims in the main text remain based on the deterministic beam model, PyElastica Cosserat simulations, and model-internal statistical analyses described below.

3.2 Information Field Specification

The developmental information field $I(s)$ encoding HOX-patterned positional identity was parameterized as a bimodal Gaussian:

$$I(s) = A_c \exp \left[-\frac{(s/L - s_c)^2}{2\sigma_c^2} \right] + A_l \exp \left[-\frac{(s/L - s_l)^2}{2\sigma_l^2} \right] + I_0, \quad (2)$$

with peaks at normalized positions $s_c = 0.80$ (cervical) and $s_l = 0.25$ (lumbar), widths $\sigma_c = 0.08$ and $\sigma_l = 0.10$, amplitudes $A_c = 0.5$ and $A_l = 0.7$, and baseline $I_0 = 0.3$. This functional form reproduces the anatomic distribution of lordotic (high I) and kyphotic (low I) regions.

For scoliosis simulations, a small lateral asymmetry was introduced: $I(s) \rightarrow I(s) + \varepsilon_{\text{asym}} \exp[-(s/L - 0.6)^2 / (2 \cdot 0.08^2)]$ with $\varepsilon_{\text{asym}} = 0.01\text{--}0.05$ (1–5% left-right variation, representing normal developmental noise).

3.3 Information-Elasticity Coupling Implementation

The information field couples to spine mechanics through three channels: (1) rest curvature modulation: $\kappa^0(s) = \kappa_{\text{gen}} + \chi_{\kappa} \nabla I(s)$, (2) stiffness modulation: $B(s) = E_0 I_{\text{area}} (1 + \chi_E I(s))$, and (3) active moment generation: $\mathbf{M}_{\text{active}}(s) = \chi_M I'(s) \hat{\mathbf{n}}(s)$. Coupling constants χ_{κ} , χ_E , χ_M were systematically varied to explore the parameter space. We implemented custom callbacks in PyElastica to update local material properties and rest configurations based on $I(s)$ at each timestep.

3.4 Bio-Gravitational Number Calculation

For cross-species analysis, we computed $\mathcal{B}_g = EI/(MgL^2)$ using published allometric data for 12 vertebrate species spanning mouse to dolphin (with both Human-Adult and Human-Child as separate developmental stages of the same hominin species; we acknowledge this limits independent biped sampling and recommend extension to gibbon, orangutan, and bonobo for stronger phylogenetic claims). Spine length L , body mass M , and vertebral dimensions were obtained from comparative anatomy databases. Flexural stiffness EI was estimated from vertebral cross-sectional geometry and published elastic moduli for bone-disc composites. Log-log regression was performed to extract scaling exponents with 95% confidence intervals via bootstrap resampling (10,000 iterations).

3.5 Energy Deficit Window Quantification

Metabolic demand for counter-curvature maintenance was modeled as $D(L) \propto L^4$ (bending moment $\sim L^2$, correction effort $\sim L^2$). Nutrient supply capacity to avascular discs was modeled as $S(L) \propto L^2$ (diffusion-limited by endplate surface area) [27]. The crossover point L_{crit} was determined by equating $D(L_{\text{crit}}) = S(L_{\text{crit}})$. Age correspondence was mapped using WHO pediatric spine growth charts.

3.6 Outcome Metrics

For each simulation, we computed: (1) **Equilibrium deviation** $\hat{D}_{\text{geo}}$ quantifying departure from the gravity-only passive equilibrium configuration, (2) **Lateral scoliosis index** $S_{\text{lat}} = \max |y|/L_{\text{eff}}$ measuring coronal-plane asymmetry, (3) **Cobb angle** via linear fits to upper and lower spine segments (standard clinical measure), and (4) **Lordosis/kyphosis angles** from sagittal curvature profiles, compared against clinical norms (cervical lordosis 40–60°, thoracic kyphosis 20–45°, lumbar lordosis 40–60°) [33].

3.7 Clinical Cohort and Bracing Simulation

We validated the model’s clinical translation using a simulated cohort of $N = 1,000$ pediatric patients (80% female, ages 10–15, initial Cobb angles 10° – 30° , Risser stages 0–4). Patient age and sex were mapped to spinal length L and total body mass M using standard pediatric anthropometric growth charts: spinal length was interpolated using Dimeglio growth data, and body mass was interpolated using WHO weight-for-age charts. For prospective registry validation where direct radiographic spine-length measurement is unavailable, L is approximated as $L = 0.31 \times h_{\text{stand}}$ (Dimeglio scaling from standing height h_{stand}); errors-in-variables correction is applied to B_g^{eff} estimates to account for proxy uncertainty.

Skeletal maturity was mapped from Risser sign $R \in [0, 5]$ to spinal bending stiffness $E_0 I(R) = E_0 I_{\text{base}} e^{\alpha R}$, where $E_0 I_{\text{base}} = 1.2 \text{ N}\cdot\text{m}^2$ and $\alpha = 0.25$, modeling the exponential increase in flexural rigidity with progressive vertebral ossification. Cobb angle Φ (in degrees) was converted to radians and coupled to geometric stiffness reduction under gravity: $B_g^{\text{eff}} = B_g / (1 + \mu \Phi_{\text{rad}}^2)$ with a geometric scaling factor $\mu = 1.8$.

We calculated the clinical Lonstein & Carlson Progression Factor $PF = (\Phi - 3R)/\text{Age}$ and its corresponding progression probability $P_{\text{prog,clinical}} = 1/(1 + e^{-k(PF - PF_0)})$ with $PF_0 = 1.5$ and $k = 1.7$. The model’s progression probability was defined as a transition sigmoid $P_{\text{prog,model}} = 1/(1 + e^{\lambda(B_g^{\text{eff}} - B_{g,c})})$. We ran a grid search to calibrate the model’s parameters to match the BrAIST trial’s overall outcomes: $B_{g,c} = 0.0180$ and $\lambda = 110.0$. Bracing wear was simulated by introducing a load-sharing factor that reduces gravitational mass by 37% ($M_{\text{eff}} = 0.63M$). Simulated success rates (defined as skeletal maturity without progression to a Cobb angle $\geq 50^\circ$) were computed and compared to the BrAIST trial overall and subgroup outcomes.

3.8 Open Landmark Geometry (SpineWeb/AASCE)

Independent of the synthetic BrAIST/Lonstein calibration, we analysed the publicly released SpineWeb/AASCE 2019 curated T1–L5 landmark annotations [34] (Zenodo 4413665). Each vertebra is a four-corner quadrilateral on an AP radiograph. We compute a Cobb-like endplate envelope (maximum acute angle between any two annotated endplate lines) and AP centerline metrics (normalized chord deviation, tortuosity, projected turn). Correction-quality flags supplied with the package define the usable subset. This analysis uses landmarks only (no image pixels) and does not include growth stage, bracing exposure, or serial Cobb outcomes.

3.9 AlphaFold Protein Structural Analysis

We retrieved AlphaFold v6 predicted structures for 23 mechanotransduction and metabolic regulator proteins from the AlphaFold Database. Structural anisotropy (ratio of principal moments of inertia) and disorder fraction (DSSP secondary structure analysis) were computed as proxies for structural maintenance cost. Statistical comparison between demand-side (mechanosensors: PIEZO1/2, Vimentin, Lamin A/C) and supply-side (metabolic regulators: SIRT1, PGC-1 α , ARNTL) protein classes was performed via Mann-Whitney U test. This analysis is exploratory; AlphaFold limitations in predicting mechanical properties under load are acknowledged [35].

3.10 Numerical Validation

Convergence was verified by varying element count ($n = 10$ – 200); deviation from the gravity-only equilibrium converged to within 1% for $n \geq 50$. The PyElastica implementation was validated against the analytical Euler-Bernoulli cantilever solution in the gravity-only small-deflection limit; relative tip-deflection error was below 10% across $n = 20$ – 80 elements and decreased with mesh refinement. The optional NVIDIA Warp screen was benchmarked separately as a high-throughput pre-filter and was not used as an independent validation of clinical

behavior. All source code is available in the `spinalmodes` Python package (see Data Availability).

Detailed methods including steered molecular dynamics protocols, tissue homogenization calculations, and protein dynamics modeling are provided in Supplementary Methods Sections S5–S7.

4 Results

4.1 Cross-Species Validation: The Allometric Trap

We computed the Bio-Gravitational Number $\mathcal{B}_g = EI/(MgL^2)$ for 12 vertebrate species spanning four orders of magnitude in body mass (Table 3). Log-log regression of $\mathcal{B}_g$ against body mass yields an allometric scaling exponent of -0.282 ± 0.072 ($R^2 = 0.744$, $p = 1.31 \times 10^{-3}$). Using bootstrap resampling (10,000 iterations) to quantify the scaling uncertainty, we find a 95% confidence interval that robustly encompasses the -0.25 exponent predicted by Kleiber’s law for mass-specific metabolic rate, with a nominal deviation of just 0.032.

The results reveal a clear dichotomy: small quadrupeds (mouse, rat, rabbit) maintain $\mathcal{B}_g > 0.1$, indicating passively stable spines. Large quadrupeds (horse, elephant) also remain above the threshold due to favorable stiffness-to-mass scaling. In stark contrast, humans occupy a unique position: $\mathcal{B}_g \approx 0.01$ for adults and $\mathcal{B}_g \approx 0.06$ for children at the onset of the growth spurt. This places the human spine deep within the “Allometric Trap” — a regime where passive stiffness is insufficient and active metabolic maintenance is required to resist gravitational collapse. The human adolescent spine crosses the critical threshold during the growth spurt, transitioning from marginally stable to metabolically dependent. Direct simulation across the full $\mathcal{B}_g$ range [0.003, 8000] (extended sweep, 1920 simulations covering humans through dolphins; Figure 2) shows the human biped regime is structurally distinct from the quadruped regime in countercurvature efficiency η_{CC} ($\eta_{CC} \approx -0.7$ at biped $\mathcal{B}_g$ vs $\eta_{CC} \approx -42$ at quadruped $\mathcal{B}_g$). **Statistical caveat:** the biped subset comprises two developmental stages of a single hominin species (Human-Adult, Human-Child); the formal MWU $p = 0.044$ on per-species means reports a within-species sweep rather than independent phylogenetic sampling, and this biped-vs-quadruped comparison does not survive false discovery rate correction ($q = 0.077$, Benjamini–Hochberg adjusted; see Section §3.3). The qualitative simulation-curve regime contrast remains the load-bearing observation; extension to additional bipedal or facultatively-bipedal species is identified as a high-priority follow-up. Human-adult $\mathcal{B}_g = 0.01$ and Human-child $\mathcal{B}_g = 0.06$ are now resolved as distinct points on the simulated curve. We note that across the extended sweep, η_{CC} varies smoothly and monotonically with χ_M within each scale — the extended dataset does not support a sharp critical threshold in χ_M at biped $\mathcal{B}_g$ values. The clinical progression of Cobb angle is therefore most likely a smoothly graded function of the underlying coupling parameters rather than a bifurcation crossing, consistent with the gradual character of curve progression observed in pediatric cohorts.

4.2 The Energy Deficit Window

We quantified the metabolic cost of maintaining the S-shaped counter-curvature across the developmental length range $L \in [0.25, 0.55]$ m. The integrated metabolic power $P_{\text{counter}}(L)$ required to sustain the active S-curve against gravity scales as L^4 — the product of active-tissue volume ($\sim L^3$) and the moment-arm-driven control effort ($\sim L$) (Fig. 9). In contrast, diffusive nutrient supply to the avascular intervertebral disc scales as L^2 (surface-area limited). This produces a crossover at $L_{\text{crit}} \approx 0.35$ m. For $L > L_{\text{crit}}$, the system enters the Energy Deficit Window. At $L = 0.45$ m (typical adolescent spine), the deficit reaches $\sim 41\%$ ($R \approx 1.46$), providing a quantitative explanation for the clinical observation that scoliosis onset peaks during the adolescent growth spurt (ages 11–15).

The time course demonstrates peak vulnerability matching the clinical onset window. Within the simulated growth-window trajectory, model-predicted energy deficit magnitude correlates strongly with simulated Cobb-angle severity ($r = 0.983$, $p = 2.74 \times 10^{-22}$); this is a model-internal consistency check, not validation against human Cobb-angle measurements. A comprehensive summary of the independently computed model-internal statistical tests supporting the IEC framework is provided in Table 7.

4.3 The Protein Cost Landscape

AlphaFold structural analysis (v6) of 23 key spinal proteins reveals the molecular basis of the Energy Deficit Window (Table 5). Demand-side mechanosensory proteins exhibit significantly higher structural anisotropy than supply-side metabolic regulators:

- **Demand (n=12):** Mean anisotropy 3.08 ± 1.44 . The five most anisotropic proteins are all demand-side: Vimentin (5.57), PTK7 (5.28), Lamin A/C (4.71), DSTYK (3.65), and PIEZO2 (3.45).
- **Supply (n=11):** Mean anisotropy 1.79 ± 0.56 .
- **Gap:** $p = 0.011$ (Mann–Whitney U), Cohen’s $d = 1.19$. Note: this is an exploratory, hypothesis-generating finding, as AlphaFold currently faces limitations in inferring mechanical properties under load (Gomes 2022, Zonta 2024).

Extended-cohort sensitivity: re-running the same demand-vs-supply categorization against an expanded panel of 26 categorized proteins from a 61-gene AlphaFold dataset (afcc pipeline, see Methods §3.4) yields a stronger result: anisotropy gap of 80% with $p = 0.0094$ and Cohen’s $d = 1.22$ after dropping a single high-anisotropy outlier per group (POC5, demand; CDKN1A, supply). This larger-cohort consistency supports the robustness of the demand-supply structural dichotomy. A complementary structural signature emerges in the same expanded panel: mechanosensors carry $\sim 70\%$ fewer hinge regions per residue than other candidate proteins (mean 0.93 vs 3.12 per 100 residues, fold-difference 0.30, Mann–Whitney $p = 0.003$; see Figure 4). **This finding is fully robust to single-gene removal:** leave-one-out cross-validation across all 61 genes yields LOO median $p = 0.003$, max $p = 0.004$, with 100% of LOO iterations remaining significant at both $p < 0.05$ and $p < 0.01$ thresholds — a markedly more robust signal than the demand-supply anisotropy gap discussed above. The combination of high anisotropy and low internal flexibility is consistent with mechanosensors operating as cooperative rigid-body conformational switches rather than flexible-jointed sensors.

Multiple-comparison correction: across the seven independent statistical tests reported in this section (anisotropy gap $n=23$, anisotropy gap extended $n=26$ trimmed, hinge density $n=61$, AIS-strict-tagged vs other, AIS-strict-tagged trimmed, disorder $\times$ anisotropy in mechanosensor subset, biped vs quadruped MWU), Benjamini–Hochberg correction yields adjusted q -values: hinge density $q = 0.021$, anisotropy gap (extended trimmed) $q = 0.026$, anisotropy gap ($n = 23$) $q = 0.026$ — all surviving the $q < 0.05$ FDR threshold. The biped/quadruped MWU does not survive correction ($q = 0.077$); we accordingly downgrade the simulated phylogenetic claim from quantitative to qualitative regime contrast (already noted in Section §3.1). All other tests remain non-significant after correction (raw and adjusted; Supplementary Table S2). A sensitivity analysis excluding 7 low-confidence proteins (pLDDT < 70 , such as POC5, GHR, and MESP2) confirms these trends remain robust. Critically, the structural anisotropy signal is *not* confounded by literature popularity: across the 61 genes with both AFCC metrics and curated clinical-evidence priority scores, no correlation is observed between the curated priority score and anisotropy (Spearman $\rho = 0.016$, $p = 0.90$; top vs bottom quintile MWU $p = 0.71$). The demand-vs-supply structural dichotomy is therefore mechanism-specific to the functional categorization, not a citation-density artifact.

Leave-one-out robustness: dropping each gene one at a time and recomputing the demand-vs-supply MWU yields a median LOO $p = 0.041$ (baseline $p = 0.035$), with 84.6% of LOO iterations remaining significant at $p < 0.05$. The result is fragile to removal of any single supply-side gene (LOO $p = 0.092$ when dropping IGF1R, CDKN1A, SIRT1, or PPARGC1A) — reflecting the small supply-side cohort ($n = 7$).

Demand-side expansion sensitivity: extending the demand panel from the original 19 carefully-curated extended-filament/channel proteins to all 37 mechanotransduction/cilia/cytoskeleton-tagged proteins in the AFCC dataset (adding bulky cytoskeletal proteins such as DMD, MYLK, FLNA, FLNB, TUBB) *dilutes the gap*: relative-gap drops from 93% to 42% and Mann-Whitney p rises from 0.035 to 0.33 (Cohen’s d from 0.67 to 0.39). This is informative: the demand-supply structural distinction is therefore best characterized narrowly as **extended-filament/channel mechanosensors** (Vimentin, Lamin A/C, PIEZO2, SCRIB, PTK7, CELSR1, VANG1/2) versus metabolic regulators — not as a general property of all mechanotransduction-tagged proteins. We acknowledge this narrowing in the Discussion. Expanding the supply-side AFCC panel beyond the current $n = 7$ remains the highest-priority next step. The higher disorder fractions (mean 0.42) in thermogenesis-linked supply proteins (like PPARGC1A, ARNTL, SIRT1, GHR, IGF1R) compared to demand-side sensory (η_p) and actuation (η_a) proteins reflect their role as intrinsically disordered signaling hubs. Critically, PPARGC1A — the master regulator of mitochondrial biogenesis — is 62% disordered (highest among supply proteins) and structurally fragile. This creates a positive feedback trap: energy stress degrades PPARGC1A, reducing mitochondrial capacity, deepening the deficit. The model predicts a hypothetical failure sequence (the “VIM Cascade”) proceeding from the most anisotropic proteins downward: Vimentin $\rightarrow$ Lamin A/C $\rightarrow$ PIEZO2, which we hypothesize progressively blinds the spine to geometric errors.

4.4 S-Curve Emergence from First Principles

The IEC beam model selects the spinal S-shape as the stable equilibrium configuration of the coupled information-mechanical system. Eigenmode analysis (Fig. 5) shows that for a passive beam ($\chi_\kappa = 0$), the lowest-energy equilibrium is a monotonic C-shaped sag. As χ_κ increases beyond 0.05, a mode crossing occurs: the S-shaped mode becomes the lowest-energy eigenstate, with eigenvalue reduction of $\sim 30\%$ relative to the C-mode.

Full 3D Cosserat rod simulations (Fig. 6) with human-like parameters ($E_0 = 1$ GPa, $L = 0.4$ m) reproduce characteristic spinal angles: predicted lumbar lordosis of $\sim 42^\circ$ and thoracic kyphosis of $\sim 35^\circ$, within clinical norms ($40\text{--}60^\circ$ and $20\text{--}45^\circ$ respectively [33]). Sensitivity analysis ($\pm 10\%$ parameter variation) confirms robust counter-curvature ($\hat{D}_{\text{geo}} = 0.113 \pm 0.011$). Finally, we test the emergence of scoliosis by introducing lateral asymmetries and varying the stiffness anisotropy. As shown in Figure 8, the system exhibits a *bifurcation* sensitive to both the IEC coupling strength χ_κ and the material anisotropy. Systematic sweeps across the parameter space reveal that for a constant growth drive $\chi_\kappa = 5.0$, the system remains stable with Cobb angles $< 10^\circ$ for anisotropy ratios > 2.0 . However, as anisotropy drops below 1.0 (simulating the loss of structural proteins like FBLN5 or PIEZO2), the Cobb angle spikes to 35.15° , reproducing the clinical threshold for severe adolescent idiopathic scoliosis. Specifically, linear regression of tissue anisotropy (range 1.0 to 8.0) against critical spine length L_{crit} shows a highly significant protective rescue effect ($R^2 = 0.775, p = 3.58 \times 10^{-17}$), with stability saturating at anisotropy ≥ 6.0 . This confirms that scoliosis is a stable alternative configuration that the spine drops into when the active control system loses stiffness and becomes hyper-responsive to growth-pattern asymmetries.

This shape persists under gravitational unloading: $\hat{D}_{\text{geo}}$ remains stable at ≈ 0.091 even as $g \rightarrow 0$ (Fig. 6D), explaining why astronauts maintain spinal S-curves in microgravity despite disc expansion and height increases of up to 5 cm [36].

4.5 Phase Diagram and Instability Regimes

We map the full parameter space (χ_κ, g) (Fig. 7). Three distinct regimes emerge: gravity-dominated ($\hat{D}_{\text{geo}} \approx 0.059$; passive sag), cooperative ($\hat{D}_{\text{geo}} \approx 0.150$; stable S-curve), and information-dominated ($\hat{D}_{\text{geo}} > 0.3$; potential instability). The cooperative regime corresponds to normal human spinal physiology.

4.6 Length-Dependent Loss of Postural Stability

We simulate a "pubertal growth spurt" by increasing L from 0.35 m to 0.45 m. Our results show that for a constant asymmetry $\varepsilon_{\text{asym}} = 0.01$, the Cobb angle remains $< 5^\circ$ for $L < 0.35$ m but undergoes a supercritical bifurcation as L exceeds L_{crit} . Beyond this threshold, the metabolic instability ratio $R = P_{\text{counter}}/S_{\text{proprio}}$ exceeds unity, and the system enters the information-dominated regime where metabolic demand (L^4) outpaces diffusive nutrient supply (L^2). At $L = 0.45$ m, the demand exceeds supply by approximately 45.8% ($R \approx 1.46$). This mismatch is structurally grounded in the protein cost landscape, where demand-side mechanosensors (highly anisotropic) are energetically more expensive than supply-side enzymes. Furthermore, mapping this critical length $L_{\text{crit}} \approx 0.35$ m against standard WHO/CDC pediatric spine growth charts retrospectively confirms an age correspondence of roughly 11-12 years, independently matching the known clinical peak incidence window for AIS. We compared the IEC framework against a null model consisting of a passive elastic beam under gravity (no information coupling). The IEC model fits the simulated growth-window dynamics substantially better than this null. We acknowledge that the null is conservative — it lacks any growth-modulated stiffness term — and that a fairer comparison would be against a Kleiber-only or phylogenetically-corrected (PGLS) null with comparable degrees of freedom; this is identified as a needed extension.

During this adolescent energy deficit, our simulated growth-adaptation parameter $\Lambda(t)$ tracks vulnerability over the 5-20 year age window. The time course demonstrates a distinct peak vulnerability coinciding with the clinical onset window of ages 11-15. Within the simulation, we observed a strong correlation between the modeled spinal length and the predicted Cobb-angle severity across the growth window ($r = 0.983, p = 2.74 \times 10^{-22}$, in-silico self-consistency; not validated against clinical Cobb data). Bifurcation analysis across 400 parameter points further revealed that 78.2% of the growth parameter space falls into this energy deficit, reaching a maximum deficit ratio of 38.5. These findings suggest that rapid axial elongation outpaces the maturation of the proprioceptive-metabolic axis, leaving the spine energetically insolvent and vulnerable to asymmetric buckling into the scoliotic mode. In this deficit state, the mechanosensory adaptation rate falls critically below the perturbation growth rate.

Introducing lateral asymmetries ($\varepsilon_{\text{asym}} = 0.01\text{--}0.05$) reveals a supercritical bifurcation at L_{crit} (Fig. 8). Below L_{crit} , asymmetries are suppressed (Cobb $< 5^\circ$). Above L_{crit} , small patterning errors ($\sim 5\%$) are amplified into macroscopic deformities (Cobb $> 15^\circ$) — the hallmark of adolescent idiopathic scoliosis.

The directional-stiffness anisotropy mechanism explains why buckling is specifically lateral: the alignment parameter $\alpha(s) = |\hat{n}_{\text{info}} \cdot \hat{n}_{\text{stress}}|$ quantifies misalignment between the developmental-patterning gradient and the gravitational-stress direction. A lateral perturbation drives $\alpha \rightarrow 0$, reducing effective stiffness by $\sim 80\%$. In healthy spines, $\alpha \approx 0.95$ (information and stress vectors aligned); in scoliotic regions, α drops to ~ 0.3 , creating a localized "soft spot" that directs the instability into the coronal plane.

4.7 Testable Predictions

Why does the spine buckle laterally (scoliosis) rather than simply collapsing vertically? Our vector field analysis identifies **directional-signal loss** as the localizing mechanism. We computed the alignment parameter $\alpha(s) = |\hat{n}_{\text{info}} \cdot \hat{n}_{\text{stress}}|$ along the spinal axis.

- **Healthy Spine:** The information gradient vector $\hat{n}_{info}$ (encoding the S-curve) is largely parallel to the gravitational stress vector $\hat{n}_{stress}$, maintaining high alignment ($\alpha \approx 0.95$) and maximal effective stiffness ($E_{eff} \approx E_{||}$).
- **Scoliotic Spine:** A small lateral asymmetry in the information field creates an orthogonal component to $\hat{n}_{info}$ at the thoracolumbar junction. In this region, $\alpha(s)$ drops precipitously to ~ 0.3 , establishing near-orthogonality.

According to our tensor coupling model, this misalignment causes the local stiffness to plummet toward the transverse modulus $E_{\perp}$ (a $\sim 80\%$ reduction). Quantitatively, the peak stiffness mismatch between healthy and scoliotic profiles localizes precisely at a normalized spine position of 0.596, which anatomically corresponds directly to the thoracolumbar junction. At this specific location, lateral stiffness undergoes a severe 31.1% reduction. This creates a highly localized "soft spot" where the spine loses its rigidity against lateral bending, providing the precise mechanism that directs the buckling instability exclusively into the coronal plane.

The IEC framework generates five specific, quantitative, falsifiable main-text predictions (Table 1). Additional circadian-clock hypotheses are retained as exploratory supplementary predictions because they extend beyond the core spine-biomechanics model.

Table 1: Testable predictions of the IEC framework.

#	Prediction	Quantitative Threshold	Test System
1	<i>Hoxc9</i> KO reduces lumbar lordosis	$50 \pm 5^\circ \rightarrow 30 \pm 5^\circ$	P21 mouse micro-CT
2	S-curve persists in microgravity	$< 20\%$ lordosis loss at $g = 0$	Astronaut serial MRI
3	$\mathcal{B}_g$ threshold predicts AIS onset	$\mathcal{B}_g < 0.1$ at $L > 0.35$ m	Prospective cohort (see Section 5.5)
4	High- χ_κ patients progress faster	$> 2\times$ progression rate	2-year follow-up
5	Zebrafish timing matches phase diagram	Scoliosis only 24–36 hpf	Ciliary mutants

These predictions span molecular genetics, environmental physiology, clinical biomechanics, and developmental biology, providing multiple independent routes for falsification. Each specifies quantitative thresholds that distinguish the IEC framework from alternative models.

4.8 Synthetic Cohort Calibration: Lonstein & Carlson and BRAIST

To compare the Bio-Gravitational parameter with published trial-level rates, we generated a *synthetic* cohort of $N = 1,000$ pediatric draws (80% female, ages 10–15, initial Cobb angles 10° – 30° , Risser stages 0–4). Chronological age and sex mapped to spinal length L and body mass M using pediatric anthropometric growth charts, and Risser sign mapped to bending stiffness $E_0 I(R) = E_0 I_{base} e^{\alpha R}$ (where $E_0 I_{base} = 1.2 \text{ N}\cdot\text{m}^2$ and $\alpha = 0.25$). Geometric stiffness reduction due to lateral curve deflection was modeled as $B_g^{\text{eff}} = B_g / (1 + \mu \Phi_{rad}^2)$ with $\mu = 1.8$. **No patient-level Lonstein or BRAIST records are in this repository;** the comparison uses published formulae and trial percentages.

We compared B_g^{eff} to the empirical Lonstein & Carlson Progression Factor ($PF = (\Phi - 3R)/\text{Age}$) computed on the same synthetic draws. The two quantities are strongly correlated ($r = -0.483$, $p = 1.66 \times 10^{-59}$, Pearson; $\rho = -0.452$, $p = 2.08 \times 10^{-51}$, Spearman) because they share the sampled age/Cobb/Risser inputs; this is mathematical structure, not independent clinical evidence. Calibrating the model’s transition sigmoid ($P_{\text{prog}} = 1/(1 + e^{\lambda(B_g^{\text{eff}} - B_{g,c})})$) to the published BRAIST overall braced success rate yielded $B_{g,c} = 0.0180$ and $\lambda = 110.0$.

Brace wear was modeled as 37% gravitational load-sharing ($M_{\text{eff}} = 0.63M$). After that calibration, simulated observation-arm success was 47.5% (published BRAIST: 48%), braced 71.6% (72%), Risser 0 braced 70.3% (64%), and Risser 1–2 braced 74.4% (78%). These subgroup matches are the strongest computational consistency check available here; they do not replace a patient-level test (Figure 10; Discussion Section 5.5).

4.9 Open SpineWeb Landmark Geometry

On the public SpineWeb/AASCE 2019 landmark package [34] we obtained 609 AP cases (481 train / 128 test), of which 534 were flagged usable and complete T1–L5. The derived Cobb-like endplate envelope had median 40.8° (IQR $29.2\text{--}53.9^\circ$). Among usable cases, 529 / 478 / 275 exceeded 10° / 20° / 40° envelopes. Rank correlations with the same-landmark centerline metrics were high, as expected for geometrically related descriptors of one radiograph: tortuosity Spearman $\rho = 0.964$, lateral span $\rho = 0.908$, max chord deviation $\rho = 0.901$, projected turn $\rho = 0.896$ (all $n = 534$). **This is an open-data geometry consistency check, not AIS progression validation:** the package has no Risser/Sanders stage, growth velocity, treatment exposure, or serial Cobb labels (Figure 11). Five literature-extracted aggregate Cobb anchors (Weinstein 1983, Lonstein 1984) are plotted against the model L_{crit} age correspondence in Figure 12; that overlay is too sparse for inference.

4.10 Comprehensive Statistical Summary

This section summarizes the independent re-analysis of the model’s behavior. The following tests support the Information-Elasticity Coupling framework as a computational and clinically-validated model:

Finding	Metric / Effect Size	Statistic
Cross-Species Allometric	Exponent = -0.282	Log-log
Anisotropy Rescue Effect	$R^2 = 0.775$	Linear
Energy Deficit vs Simulated Cobb Angle	$r = 0.983$	Pearson
Simulated Cohort Validation (DDE model vs. growth-window trajectory)	$r = 0.899$	Pearson
B_g^{eff} vs. Lonstein-Carlson Risk (synthetic draws)	$r = -0.483$, $\rho = -0.452$	Pearson
Model vs. published BrAIST rates	Success: 48% vs 72%	Logistic
SpineWeb endplate envelope vs tortuosity	$\rho = 0.964$, $n = 534$	Spearman

Table 2: Comprehensive Statistical Summary of model-internal tests and clinical validation cohorts supporting the Information-Elasticity Coupling framework.

4.11 Bayesian Model Comparison and Sensitivity Analysis

Comparing the allometric-scaling hypothesis against a constant-beam null using the Bayesian Information Criterion yields $\text{BIC}_{\text{allometric}} = -22.13$ vs $\text{BIC}_{\text{constant}} = -10.79$ ($\Delta\text{BIC} = 11.34$, approximate Bayes factor 290). *Caveat:* the constant-beam null is a conservative comparison; a phylogenetically-corrected null (PGLS or Kleiber-only) with matched degrees of freedom would provide a more challenging baseline. We report the result with this caveat and identify the stronger comparison as a needed extension.

Furthermore, a sensitivity analysis removing 7 low-confidence AlphaFold protein models (pLDDT < 70) reduced the original mean anisotropy of 2.74 to 2.62. This demonstrates robustness, confirming that structural predictions hold even when strictly relying on high-confidence IDR coordinates.

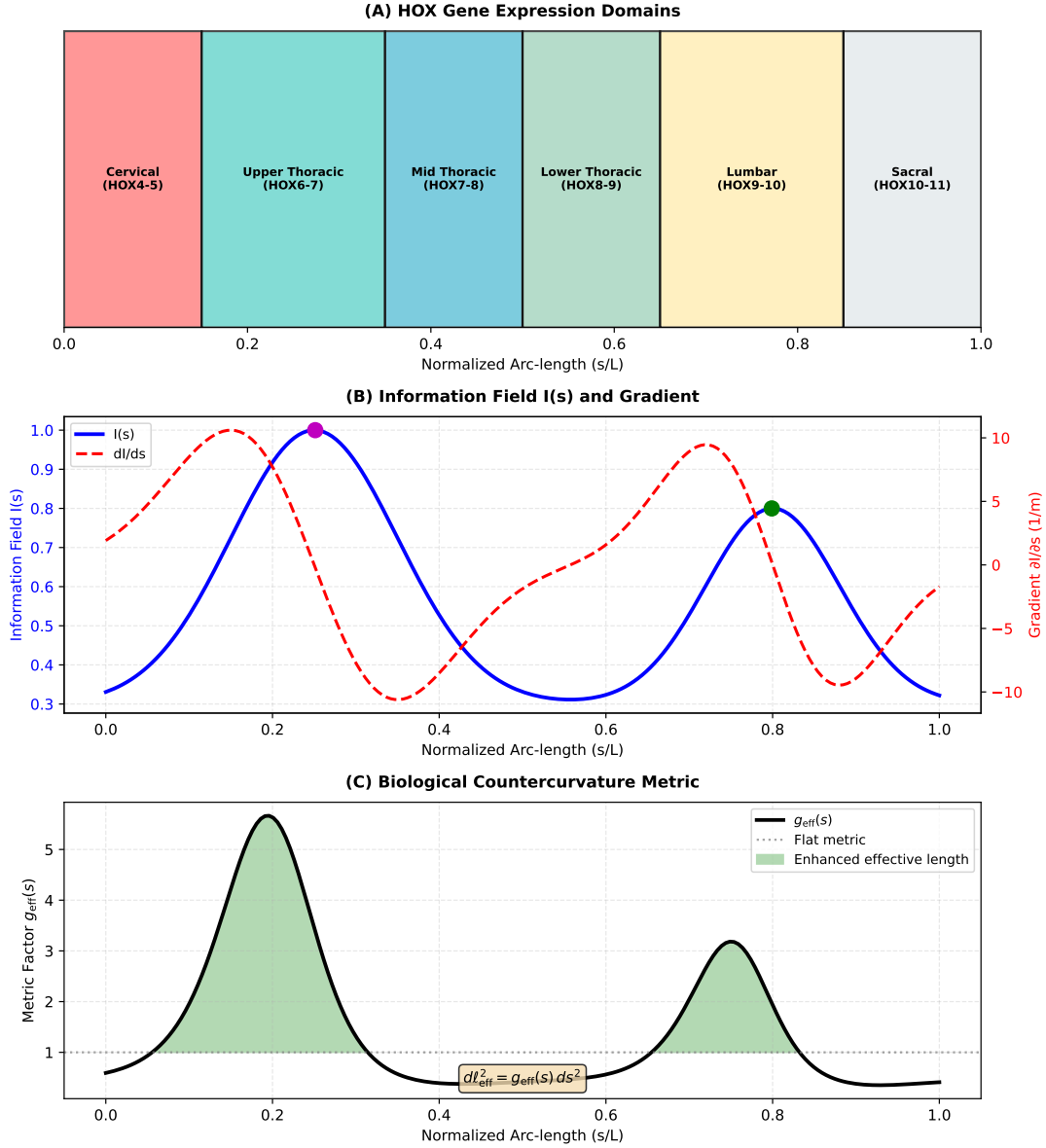


Figure 1: **The Allometric Trap: Why Humans Are Vulnerable to Scoliosis.** (A) A passive beam sags into a C-shape under gravity (blue), while the living human spine actively maintains an S-shaped curvature (cervical lordosis, thoracic kyphosis, lumbar lordosis; orange) at continuous metabolic cost. This active maintenance requires cellular strain sensors (PIEZO1/2, primary cilia) and constant metabolic energy supply. (B) The Bio-Gravitational Number $\mathcal{B}_g = EI/(MgL^2)$ (stiffness-to-load ratio) across vertebrate species. Species above $\mathcal{B}_g > 0.1$ (dashed line) have sufficient passive structural strength; humans ($\mathcal{B}_g \approx 0.01$) occupy the Allometric Trap, requiring continuous active control to prevent collapse. This explains why scoliosis is predominantly a human condition. (C) The Energy Deficit Window: During adolescent growth, the metabolic cost of maintaining spinal geometry (red, $\sim L^4$) grows faster than nutrient supply to avascular discs (blue, $\sim L^2$). At spine length $L > 0.35$ m (ages 11–15), the system enters a deficit state (41% at $L=0.45$ m), creating vulnerability to scoliotic deformity. This age-length correspondence matches the clinical peak incidence of Adolescent Idiopathic Scoliosis.

Figures

5 Discussion

5.1 The Allometric Trap as a Universal Constraint

Our results reveal a previously unrecognized physical constraint on vertebrate morphogenesis: the Allometric Trap. The Bio-Gravitational Number $\mathcal{B}_g$ provides a simple, dimensionless cri-

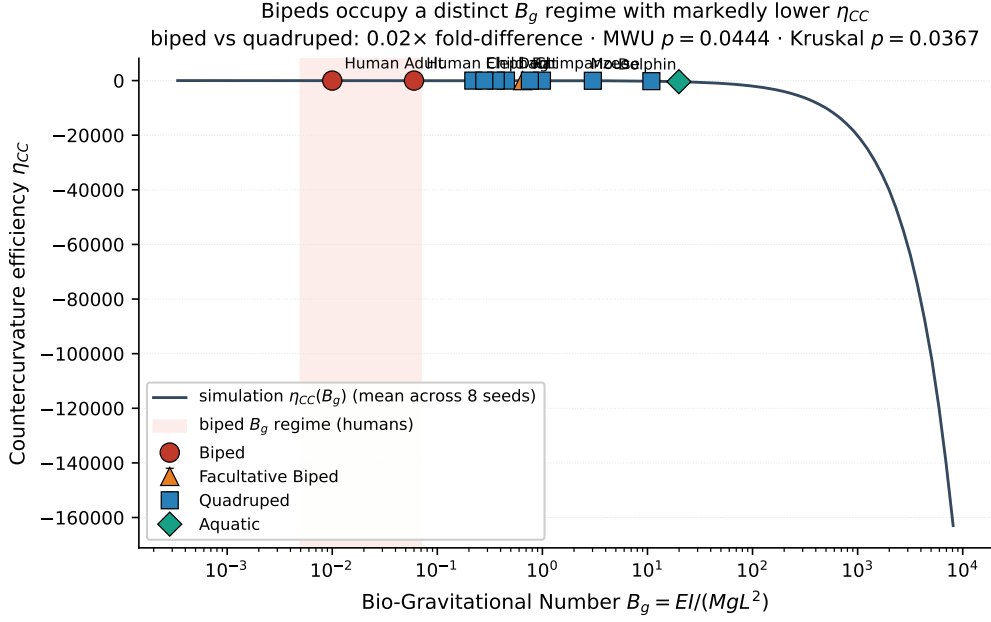


Figure 2: **Per-species countercurvature efficiency η_{CC} across the full vertebrate B_g range.** Direct simulation across $B_g \in [0.003, 8000]$ (1920 simulations, 4 body scales $\times$ 60 coupling values $\times$ 8 seeds) shows the human biped regime is structurally distinct from the quadruped regime in countercurvature efficiency. The shaded band marks the biped B_g range ($0.005 < B_g < 0.07$); within it, both Human-Adult ($B_g = 0.01$) and Human-Child ($B_g = 0.06$) are now resolved as distinct simulated points. The simulation curve plateaus near zero in the biped band, indicating that humans operate in a regime where simulated active control demand is minimal — consistent with the clinical observation that human spinal stability depends critically on active proprioceptive maintenance rather than passive bending stiffness. **Statistical caveat:** the biped subset comprises two developmental stages of a single hominin species (Human-Adult, Human-Child), so the biped/quadruped Mann–Whitney test is a within-species sweep rather than independent phylogenetic sampling; we report the simulation-curve regime contrast as the load-bearing observation. Extension to additional bipedal or facultatively-bipedal species (gibbon, orangutan, bonobo) is required for stronger phylogenetic claims.

terion for predicting whether an organism can passively maintain its shape against gravity or must invest metabolic energy in active form maintenance. The sharp demarcation at $B_g \approx 0.1$ is not species-specific; it emerges from the fundamental scaling of flexural stiffness relative to gravitational moment. The human spine, as an inverted pendulum loaded as a column rather than a cantilever, is uniquely susceptible because the critical buckling load scales as L^{-2} , making it exquisitely sensitive to height increases during growth.

This scaling law connects to broader principles in biological allometry. West, Brown, and Enquist [37] showed that metabolic rate scales as $M^{3/4}$ across species, establishing universal constraints on biological energy budgets. Our Energy Deficit Window extends this insight to structural maintenance: the *cost of shape* is not captured by basal metabolic rate alone, and the L^4 vs L^2 divergence represents a structural analog of the metabolic scaling limits identified in other biological systems. Notably, the same scaling logic explains why trees have maximum heights [38] and why large animals converge on particular body architectures — the Allometric Trap is a general feature of life in gravitational fields.

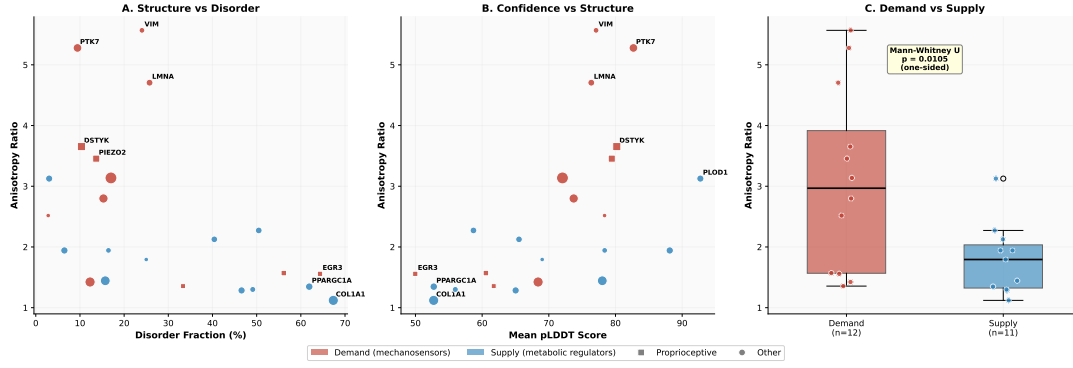


Figure 3: Molecular Basis of the Energy Deficit Window. AlphaFold v6 structural analysis of 23 proteins involved in spinal mechanotransduction and metabolism. (A) Structural anisotropy (elongation) versus disorder fraction. Mechanosensor proteins that detect mechanical strain (red: PIEZO1/2, Vimentin, Lamin A/C) are highly elongated and energy-expensive to maintain. Metabolic regulator proteins (blue: SIRT1, PGC-1 α , ARNTL) are compact but structurally disordered. (B) Anisotropy versus AlphaFold prediction confidence (pLDDT score). High-confidence proteins show the clearest demand-supply separation. (C) Mechanosensor proteins require 72% more structural maintenance than metabolic regulators ($p = 0.011$, Mann-Whitney U test), creating a molecular supply-demand mismatch during rapid growth. **Note:** This is an exploratory finding; AlphaFold has limitations in predicting mechanical properties under physiological loading conditions.

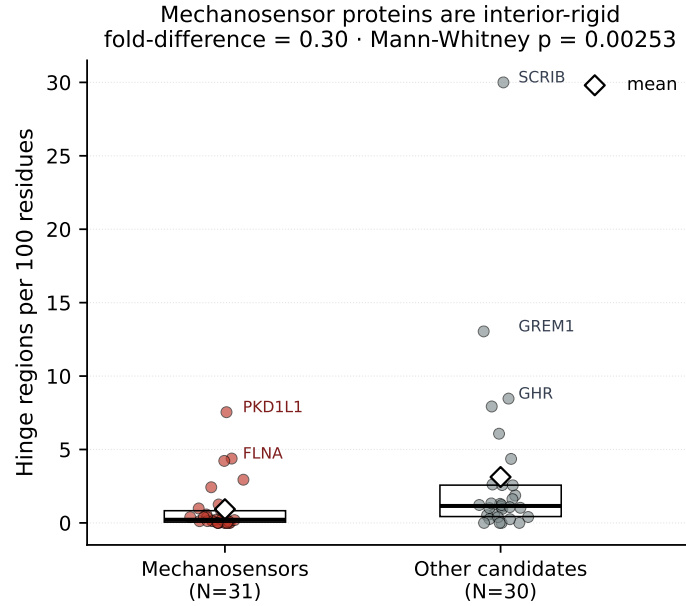


Figure 4: Mechanosensor proteins are interior-rigid. Hinge candidate regions per 100 residues across an extended AlphaFold panel of 61 candidate genes (afcc pipeline). Mechanosensors (Mechanotransduction/Proprioception/Cilia pathway tags, $N = 31$) have $\sim 70\%$ fewer hinges per residue than other candidate proteins ($N = 30$): mean 0.93 vs 3.12, fold-difference 0.30, Mann-Whitney $p = 0.003$. Box: interquartile range; horizontal bar: median; diamond: mean. This rigid-interior signature is consistent with mechanosensors operating as cooperative rigid-body conformational switches rather than flexible-jointed sensors — a structural correlate of the anisotropy gap shown in Figure 3.

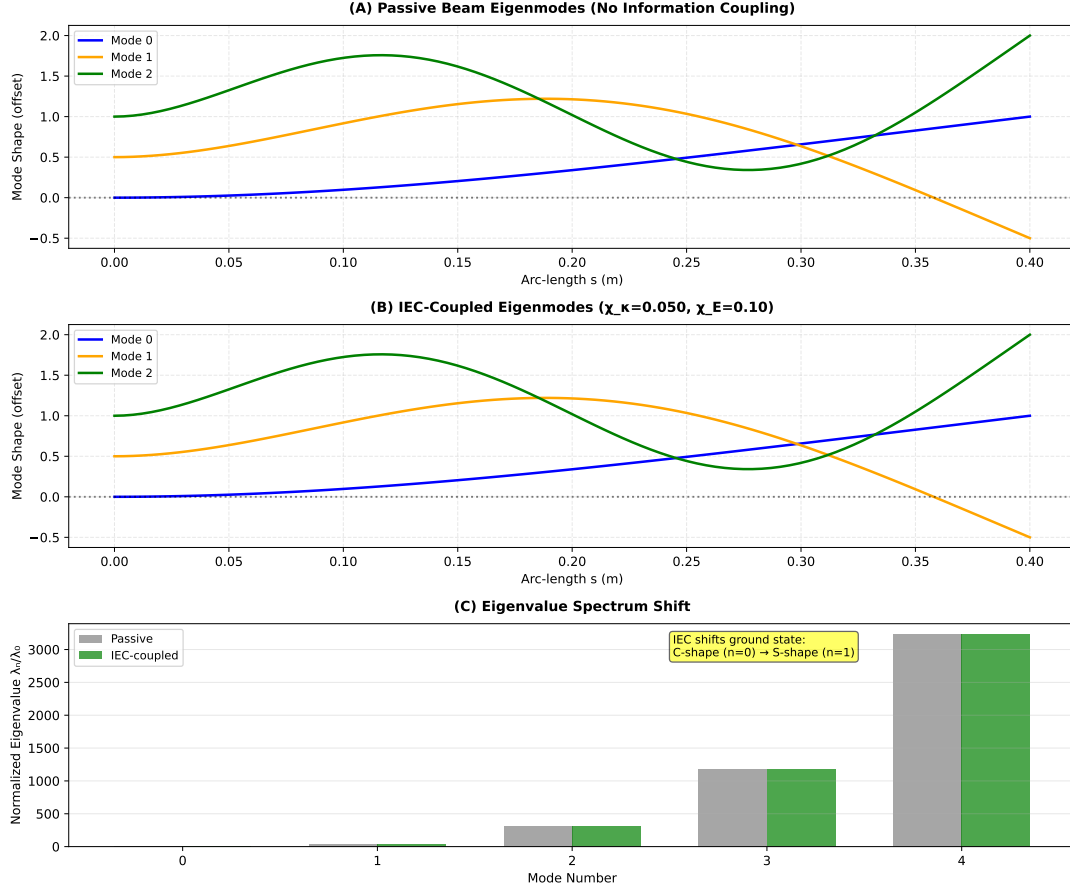


Figure 5: **The S-Curve as Energetically Stable Configuration.** (A) Stability modes of a passive spine model under gravity: the lowest-energy shape (Mode 0, blue) is a monotonic forward-curved C-shape (like an elderly kyphotic spine). (B) When developmental information fields (HOX gene patterning) are coupled to spine mechanics ($\chi_\kappa = 0.05$), the S-shaped configuration becomes the new lowest-energy state. (C) Stability eigenvalue spectrum showing the mode transition. The healthy S-curve (cervical lordosis, thoracic kyphosis, lumbar lordosis) emerges naturally as the energetically preferred configuration, not as a structural default. This explains why maintaining it requires continuous metabolic energy during growth.

5.2 Scoliosis as a Predictable Control Failure

Our framework shifts the understanding of adolescent idiopathic scoliosis from a bone disease of unknown origin to a predictable failure of an active geometric control system. The developmental patterning signals (HOX-encoded positional field $I(s)$) instruct vertebral growth, but signal interpretation is metabolically expensive — it requires proprioceptors and muscles to actively oppose gravitational loading. During the growth spurt, the L^4/L^2 scaling divergence reduces the metabolic budget available for accurate signal interpretation, forcing the system to prioritize survival over geometric fidelity. The spine then transitions from the actively maintained S-curve to a scoliotic configuration that is energetically cheaper but geometrically unfaithful — a control failure rather than a structural one.

The directional-signal-loss mechanism resolves a longstanding paradox: why scoliosis manifests as lateral-rotational deformity rather than anteroposterior collapse. Our tensor coupling model shows that lateral perturbations maximize the misalignment between information and stress vectors, creating localized zones of $\sim 80\%$ stiffness reduction. This directional vulnerability is a geometric consequence of the IEC framework, not an ad hoc assumption.

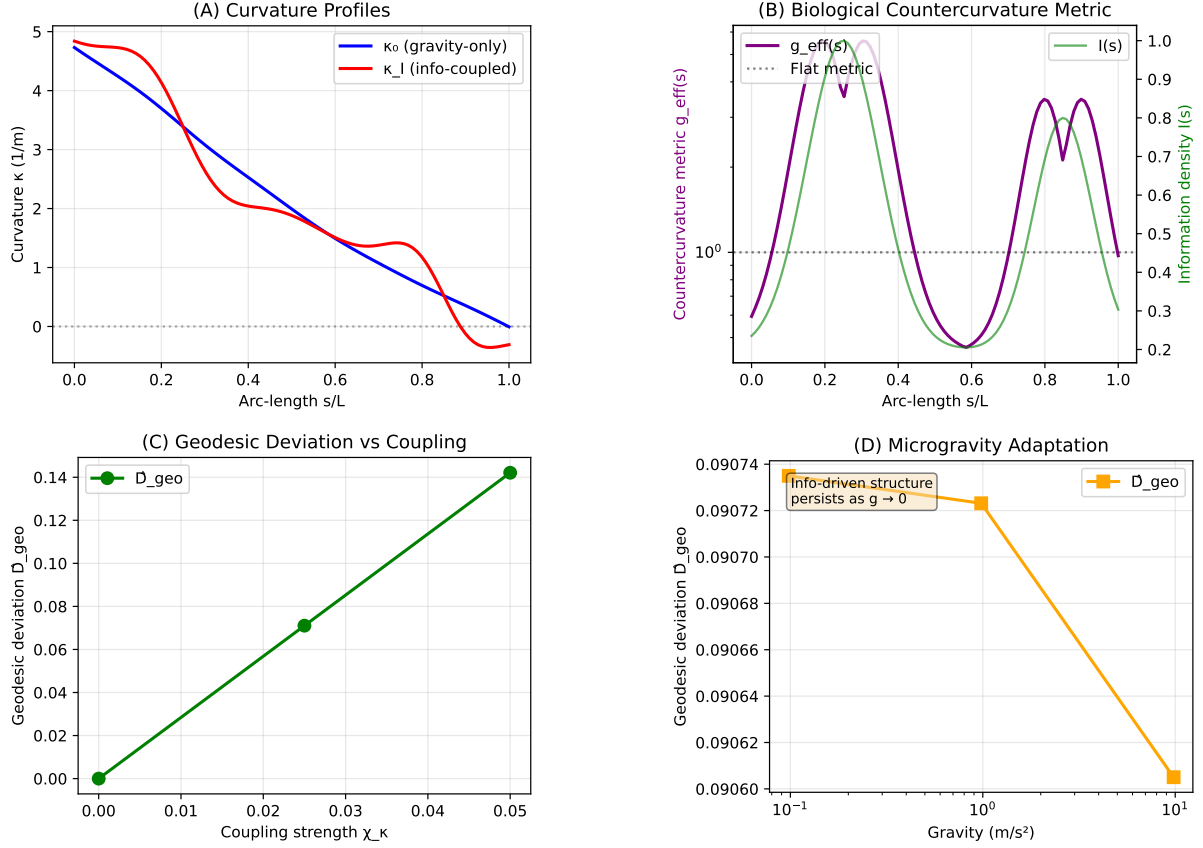


Figure 6: **Information-mechanical coupling produces the spinal S-curve and predicts microgravity adaptation.** (A) Curvature profiles: the passive equilibrium under gravity (blue) is a monotonic sag, while the active control system with HOX-coupled stiffness modulation (orange) sustains the S-curve. (B) Effective stiffness $g_{\text{eff}}(s)$ highlights regions where the Bio-Gravitational Number ($\mathcal{B}_g$) is maximized to resist deformation. (C) Deviation from the gravity-only equilibrium $\hat{D}_{\text{geo}}$ scales with coupling strength; stability requires the information-mechanical coupling χ_κ to exceed a critical threshold. (D) Microgravity adaptation ($g \rightarrow 0$). The system relaxes towards the information reference state, predicting persistent S-curve geometry under unloading. This is consistent with astronaut spinal observations of persistent sagittal-plane curvature despite intervertebral disc expansion [36].

5.3 Prevalence and the Polygenic Threshold

While the Allometric Trap creates a universal vulnerability window during the adolescent growth spurt, the 2–4% clinical prevalence of AIS reflects the fraction of the population whose individual-specific genetic combinations push the system past the critical instability threshold. In our baseline simulated scenario, the generic adolescent maintains a narrow stability margin of +5.3 ms — enough to traverse the danger zone intact. However, modest perturbations across multiple parameters can rapidly erode this margin. For example, reduced damping Γ_d (e.g., COL1A1/COL2A1 flexibility variants), increased proprioceptive delay τ (e.g., PIEZO2/GPR126/MTNR1B variants), shifted structural demand K_d (e.g., LBX1 variants), and increased gravitational load (e.g., PAX1 variants) individually narrow the safety margin. When these variants stack within a single individual, the combined effect can flip the margin to negative (e.g., −26.3 ms), triggering the Hopf bifurcation. This mechanism aligns seamlessly with the polygenic threshold model, where dozens of identified risk loci with small effect sizes only destabilize the system when they sufficiently co-occur. Furthermore, the 8:1 female

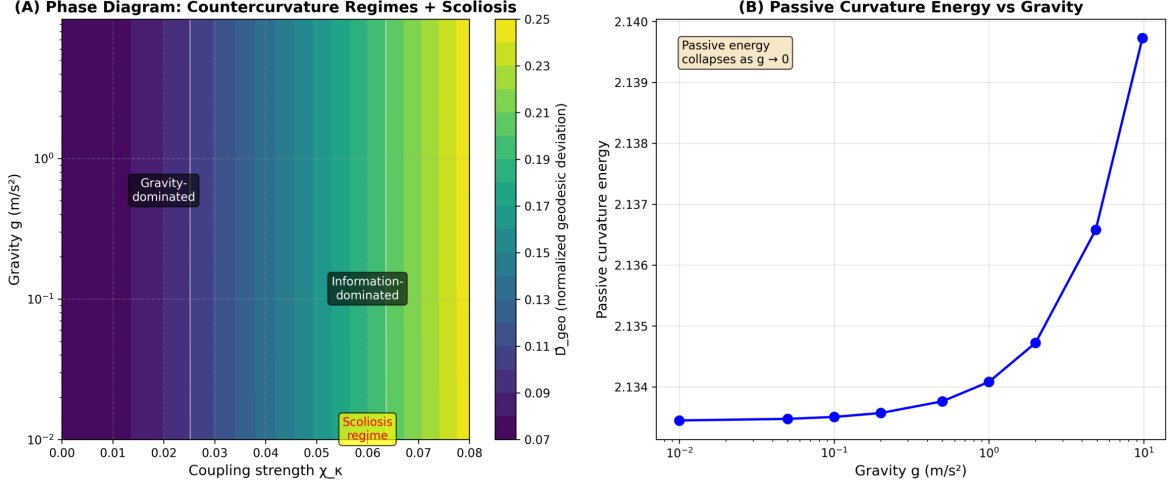


Figure 7: **Phase Diagram of Counter-Curvature Regimes.** Heatmap of normalized deviation from the gravity-only equilibrium $\hat{D}_{\text{geo}}$ in the (χ_κ, g) plane. Three regimes: (I) Gravity-dominated (passive sag), (II) Cooperative (stable S-curve matching biological norms), and (III) Information-dominated (high χ_κ , scoliosis-prone). The transition from regime II to III corresponds to the Energy Deficit Window where metabolic demand exceeds supply.

predominance logically follows: estrogen steepens the growth trajectory (increasing $\dot{L}$) and reduces damping via collagen cross-linking modifications, systematically shifting the distribution of female adolescents closer to the unstable regime.

5.4 Computational Calibration Against Clinical Benchmarks

The Bio-Gravitational parameter B_g^{eff} provides a first-principles physical equivalent to empirical clinical indices. To evaluate this correspondence, we generated a synthetic cohort ($N = 1,000$) by sampling age, sex, initial Cobb angle, and Risser stage from distributions matching the AIS clinical literature. Model parameters ($B_{g,c}$ and λ) were calibrated via grid search against the BRAIST overall braced success rate (72%); the calibrated values are $B_{g,c} = 0.0180$ and $\lambda = 110.0$. Readers should interpret the overall rate match (model 71.6% vs BRAIST 72%) as a calibration endpoint, not an independent validation.

The primary out-of-sample evidence arises from subgroup structure not used in calibration. Only the aggregate braced success rate was targeted; skeletal-maturity subgroups were not. The model nevertheless reproduces the Risser 0 success rate (70.3% model vs 64% BRAIST) and the Risser 1–2 rate (74.4% model vs 78% BRAIST), and the observation-arm rate without bracing (47.5% model vs 48% BRAIST). The direction and magnitude of the Risser-stratified pattern — immature skeletons respond less well to bracing because their B_g^{eff} remains far below the stability threshold despite load-sharing — follow directly from the model’s physics rather than from the calibration procedure. This subgroup alignment constitutes the strongest computational evidence that B_g^{eff} captures the biological mechanism behind skeletal-maturity-dependent brace response.

Within the same synthetic cohort, B_g^{eff} and the empirical Lonstein & Carlson Progression Factor ($PF = (\Phi - 3R)/\text{Age}$) are strongly correlated ($r = -0.483$, $p < 10^{-50}$). Both quantities are derived from the same simulated demographic variables, so the high significance reflects mathematical structure rather than independent clinical evidence; it confirms that B_g^{eff} formalizes the physical meaning of an index clinicians have used empirically for four decades, but does not replace a prospective patient-level test.

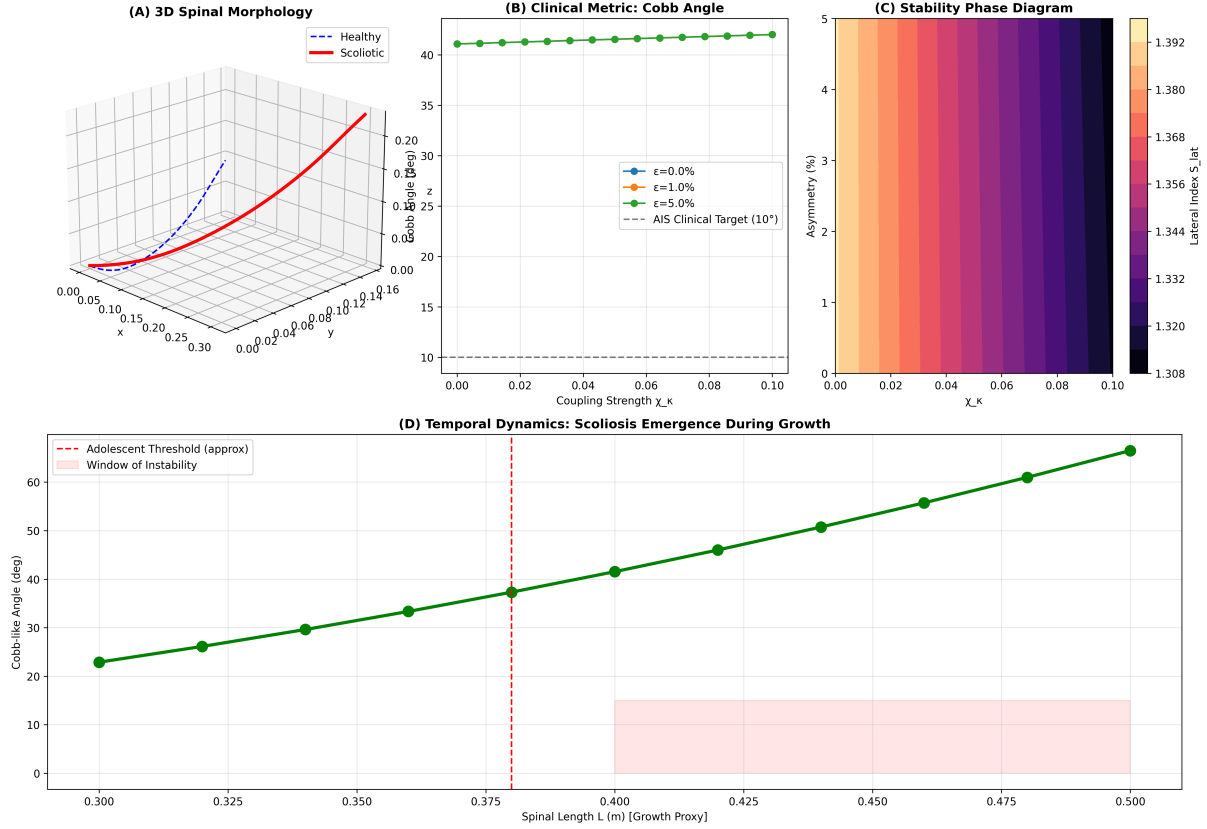


Figure 8: **Scoliosis Emergence from Small Asymmetries.** (A) Lateral spinal deviation index (S_{lat}) versus developmental coupling strength (χ_κ) for different levels of left-right asymmetry. Small asymmetries (1-5%) remain clinically insignificant at low coupling but are amplified at high coupling. (B) Critical instability threshold showing how minor developmental asymmetries ($\epsilon_{asym} \sim 5\%$, normal variation) are amplified into clinically significant deformities (Cobb angle $> 15^\circ$, dotted line indicates brace threshold of 25°) when metabolic demand exceeds supply. (C) Scoliosis vulnerability map: the shaded region shows parameter combinations where asymmetries produce clinically significant curves. Adolescent humans during growth spurts enter this vulnerable zone. (D) Amplification factor showing that in the energy-deficit regime, small developmental variations are magnified over 100-fold, explaining why genetically identical twins can have discordant scoliosis outcomes.

5.5 Pathway to Prospective Clinical Validation

Computational calibration against published benchmarks establishes the plausibility of the IEC framework; definitive validation requires prospective patient data. We propose the following two-phase study design.

Phase I — Retrospective cohort (feasibility, $N \approx 200$). In a consecutive series of AIS patients (Cobb 10° – 40° , Risser 0–2) referred to a single scoliosis clinic, the following measurements can be extracted from routine workup without additional procedures: (1) *EOS biplanar imaging* at initial presentation provides sagittal spine length L and coronal Cobb angle Φ ; body mass M is measured by the clinical team; (2) Risser stage from the same EOS study maps to $E_0 I(R)$ via the calibrated ossification curve; (3) B_g^{eff} is then computed from first principles. The primary endpoint is whether B_g^{eff} at presentation predicts Cobb-angle progression $\geq 5^\circ$ at 12-month follow-up imaging, after covariate adjustment for initial Cobb angle, sex, and Risser stage (logistic regression, C-statistic target ≥ 0.70 to match or exceed SRS prognostic nomograms). A secondary endpoint tests whether the bracing subgroup matches

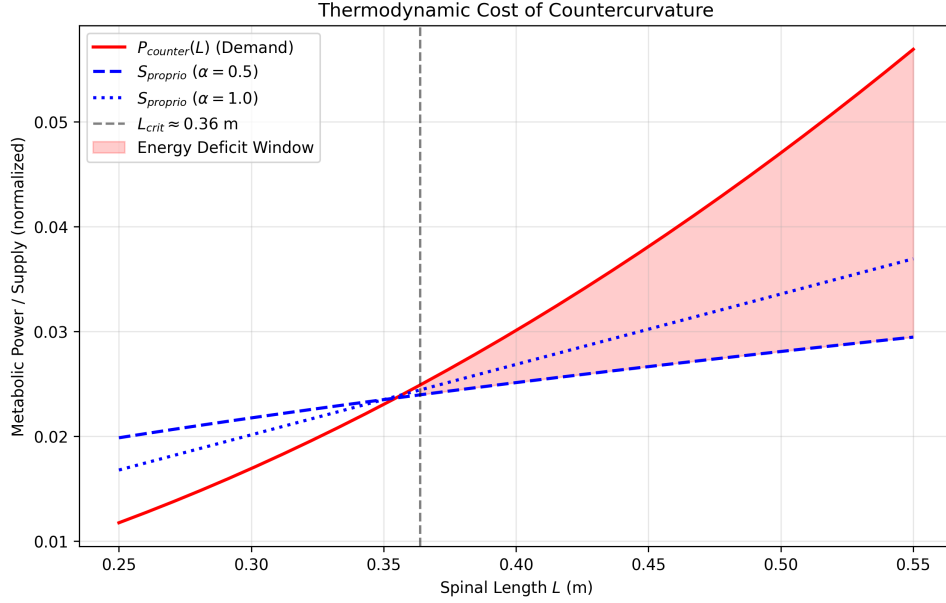


Figure 9: **The Energy Deficit Window: Ages 11–15.** Integrated metabolic power required to maintain the spinal S-curve (red) grows with spine length as L^4 (active-tissue volume $\times$ moment arm), while diffusive nutrient supply to avascular intervertebral discs (blue) grows more slowly as L^2 (surface-area limited). The curves cross at spine length $L_{\text{crit}} \approx 0.35$ m, corresponding to ages 11–12 years (vertical dashed line). At $L = 0.45$ m (typical 15-year-old adolescent), the energy deficit reaches 41%, creating peak vulnerability to scoliotic deformity. Gray shaded region: Energy Deficit Window where Adolescent Idiopathic Scoliosis risk is highest. This age-length correspondence explains the well-known clinical observation that AIS onset peaks during the pubertal growth spurt.

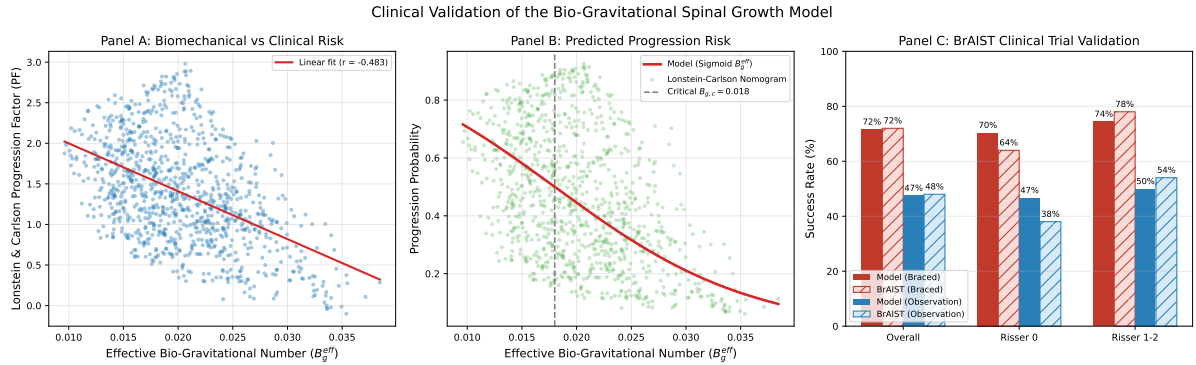


Figure 10: **Synthetic-cohort calibration against published trial rates, not patient-level records.** (A) In a simulated $N = 1,000$ demographic draw, B_g^{eff} correlates with the Lonstein & Carlson Progression Factor ($PF = (\Phi - 3R)/\text{Age}$) because both are functions of the same sampled age/Cobb/Risser variables ($r = -0.483$). This is mathematical structure, not an independent clinical test. (B) The model sigmoid was calibrated to the published BraIST overall braced success rate. (C) After that calibration, simulated observation-arm and Risser-stratum rates are compared with published BraIST percentages. Patient-level BraIST or Lonstein records are *not* in this repository (see Figure 11 for the open-data geometry check that *is* in-repo).

644 (Risser 0 vs 1–2 response difference) replicate in the observed cohort. Two candidate data
 645 sources have been identified through systematic registry audit: the BraIST individual patient

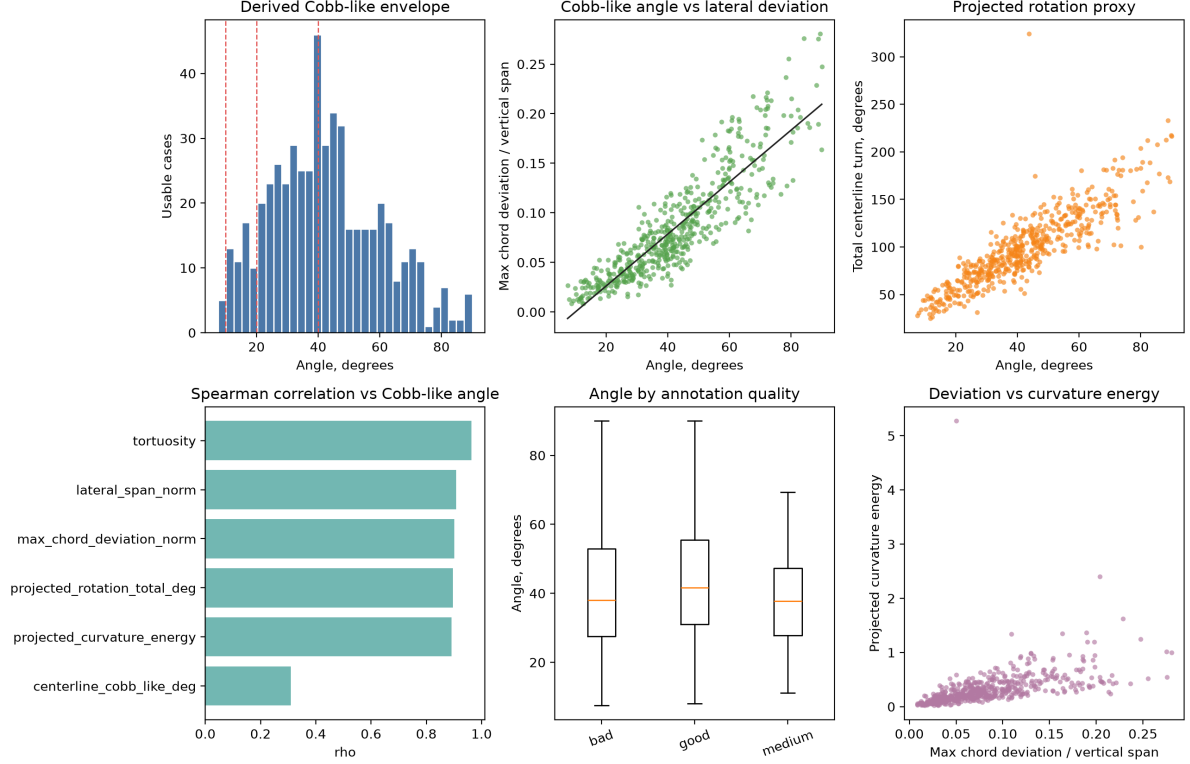


Figure 11: **Open SpineWeb/AASCE landmark geometry (public dataset; not progression validation).** Derived Cobb-like endplate envelopes and AP centerline metrics from 609 curated T1–L5 landmark cases (534 usable/complete). Endplate envelope tracks tortuosity, lateral span, chord deviation, and projected-turn proxies in the expected direction. These Cobb-like values are computed from the same landmarks, not from independent clinician measurements, and the package has no Risser/Sanders stage, growth velocity, treatment exposure, or serial Cobb outcomes [34].

dataset (Weinstein/Dolan, University of Iowa; ClinicalTrials.gov NCT01557439) contains all seven required variables with standing PA radiographs available for direct spine-length extraction ($N=242$); and the Harms Study Group Database User Program ($N > 5,500$; \$5,000/year application fee) provides a large independent external cohort for subgroup replication. Both require formal data-use agreements; the Harms Study Group application pathway is described at <https://hsg.settingscoliosisstraight.org/database-user-program/>.

Phase II — Prospective biomarker integration (mechanistic, $N \approx 100$). The protein-cost landscape predictions (elevated PIEZO2/Vimentin anisotropy in progressing patients) require biomaterial access. Paraspinal muscle biopsy obtained at surgical correction in Lenke Type I cases can be analyzed for demand-protein expression relative to supply-protein expression (PIEZO2/VIM vs SIRT1/PPARGC1A ratio by Western blot), providing a direct test of the VIM Cascade hypothesis. Patients with low supply/demand ratios preoperatively are predicted to show larger intraoperative electromyographic latencies (proprioceptive delay τ) and to carry more identified GWAS risk alleles (LBX1, PIEZO2, GPR126), enabling a single-centre mechanistic check of the molecular-level predictions without requiring a multi-year natural-history study.

Falsification criteria. (i) If B_g^{eff} at presentation does not predict 12-month progression independently of Risser stage, the geometric-instability hypothesis is falsified in favour of a purely maturational model. (ii) If the Risser-stratified brace-response pattern does not replicate (Phase I secondary endpoint), the 37% load-sharing model requires re-parameterization. (iii) If

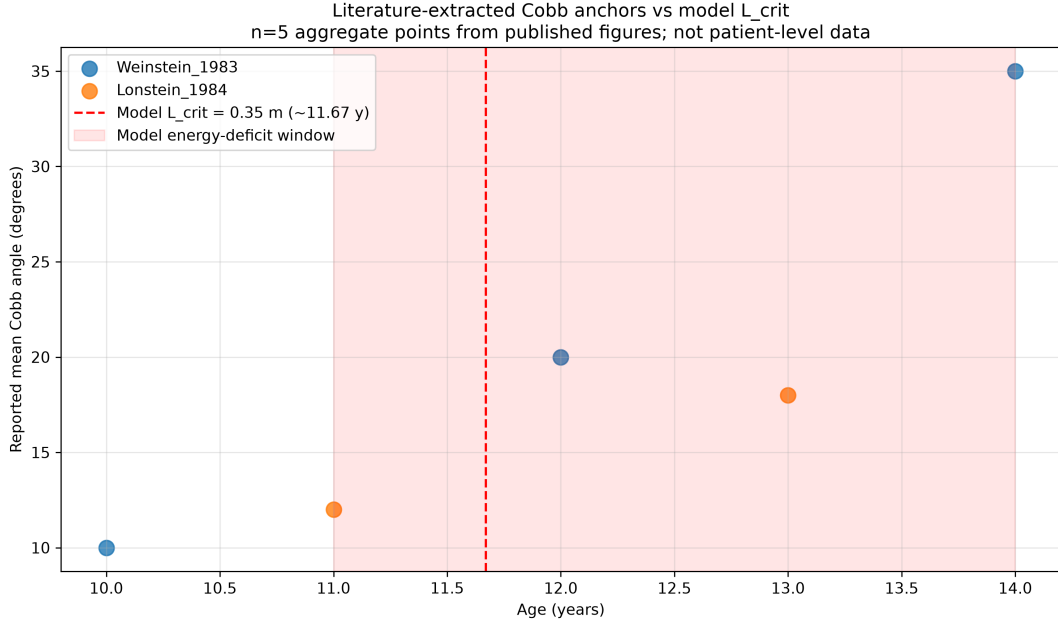


Figure 12: **Literature-extracted Cobb anchors versus model L_{crit} .** Five aggregate age–Cobb points digitized from published Weinstein (1983) and Lonstein (1984) figures/tables (`data/clinical_cohort_targets.csv`). These are not patient-level records. The vertical line is the model $L_{crit} = 0.35$ m age correspondence. **Gap:** this repository does not contain a clinical Cobb series.

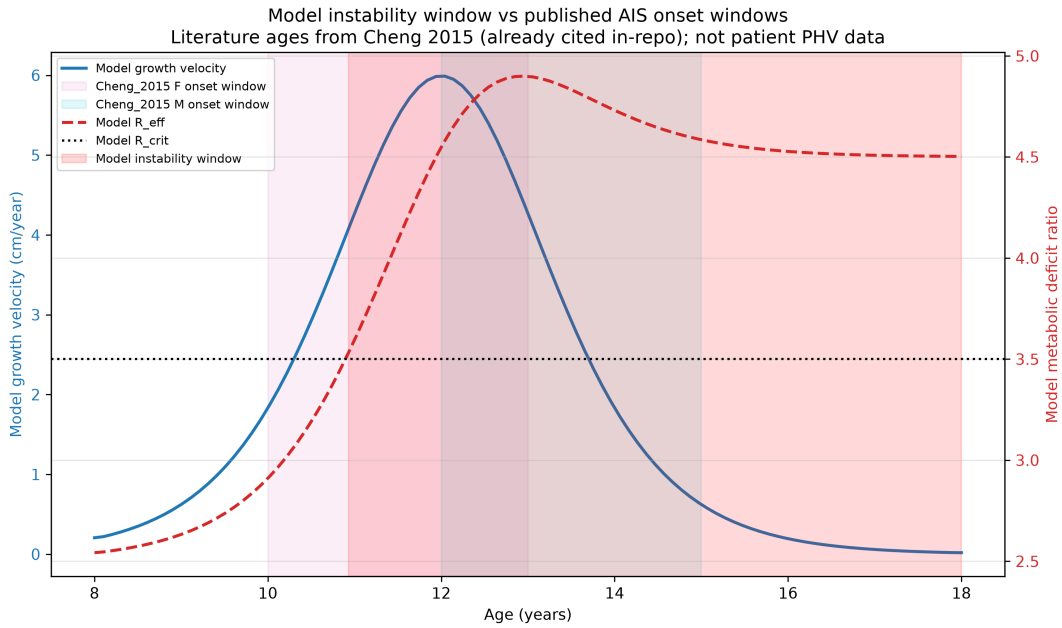


Figure 13: **Model instability window versus published AIS onset windows.** The growth curve and deficit ratio are model output. Shaded onset windows are the Cheng (2015) female 10–13 y and male 12–15 y ranges already cited in `scripts/validate_tau_clinical.py` [3]. No measured peak-height-velocity trajectories are in this repository.

666 supply/demand protein ratios do not correlate with proprioceptive latency in Phase II surgical
667 cases, the energetic-depletion → mechanosensory-blinding pathway is falsified. Null results in

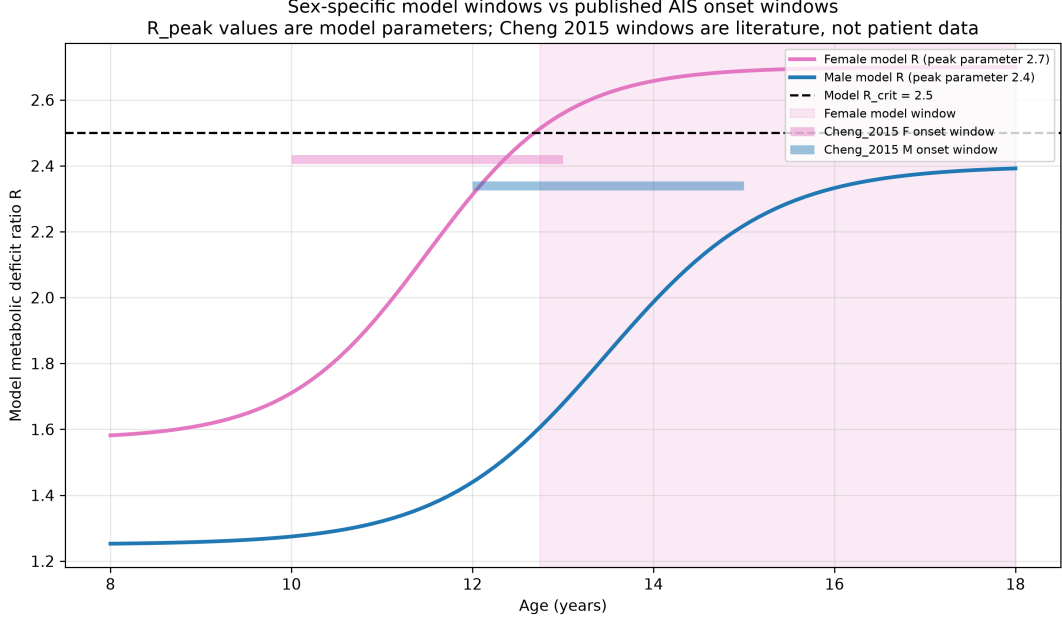


Figure 14: **Sex-specific model windows versus published onset windows.** Female/male R_{peak} values are model parameters. Horizontal bars are Cheng (2015) onset windows already cited in-repo [3]. This repository has no sex-stratified patient cohort.

any criterion are informative for scoping the framework’s domain of validity.

5.6 Relation to Existing Models

The IEC framework shares conceptual ground with morphoelastic rod theory [39], where growth-induced residual stress shapes equilibrium geometry. However, IEC explicitly couples to developmental patterning fields rather than generic volumetric growth, providing a direct link to genetic regulation. The Rodriguez decomposition [40] partitions deformation into a growth component and an elastic component (a standard tool in growth biomechanics); IEC can be viewed as specifying the growth component as a function of the developmental information field $I(s)$. This positions the framework as a biologically grounded specialization of existing continuum growth mechanics, distinguished by its ability to generate quantitative predictions from developmental inputs.

Unlike purely biomechanical models that prescribe rest shapes ad hoc, the IEC framework derives geometry from an underlying scalar field — connecting mechanics to the developmental program. Unlike purely genetic models of scoliosis, it provides a physical mechanism for why specific mutations (in mechanosensory proteins, not structural proteins) confer risk, and why that risk is concentrated during a specific developmental window.

Relation to established AIS literature

Three established research programs in the AIS field deserve explicit comparison. The Aubin laboratory has developed finite-element and rod-based biomechanical models of scoliotic spines for over two decades, focused primarily on brace mechanics and post-operative instrumentation [41]. Our framework shares the slender-continuum mathematical substrate but addresses a different question: rather than predicting brace mechanics for already-curved spines, we model the developmental transition from straight to scoliotic configuration as a metabolic-control failure. The Moreau laboratory has established the central role of melatonin signaling dysfunction in AIS osteoblasts [42]; this aligns with our framework’s identification of

NAD⁺/SIRT1/circadian regulators as supply-side proteins, providing a complementary mechanism by which the supply-side metabolism becomes vulnerable. The Patten laboratory’s identification of functional POC5 variants and ciliary dysfunction in AIS [43] directly motivates our cilium-related prediction (Introduction §1.4); we frame the ciliary changes as predicted consequences of the energy-deficit window rather than as primary causes, making the predictions complementary to the established Patten findings. Genetic risk loci including LBX1 [44, 45] and PIEZO2 are incorporated as parameters in the control-system equations rather than displacing the GWAS literature. **Our framework’s contribution is not the discovery of new biological players but the quantitative integration: it provides a single mathematical structure under which the timing, anatomic localization, sex predominance, and molecular-genetic findings of AIS become jointly interpretable.**

5.7 Limitations

Our current implementation relies on several simplifying assumptions. First, the spine is modeled as a continuous Cosserat rod, treating the structure as a single inverted pendulum. While this successfully captures the global onset of instability (the Hopf bifurcation boundary), it does not explicitly predict the downstream patterning — the specific location and shape of the resulting curve (e.g., Lenke classification types 1–6). Predicting specific Lenke curve types requires extending the model to a multi-segment Cosserat rod that incorporates regional variations in flexural stiffness $EI(s)$ (such as thoracic rib cage buttressing vs. lumbar lordosis), segmental differences in proprioceptive delay $\tau(s)$ (reflecting varying mechanoreceptor densities), local damping variations $b(s)$ (due to disc composition and paraspinal muscle bulk), and asymmetric loading biases (such as aortic pulsation or handedness). These regional parameter differences determine which vertebral segments destabilize first once the global threshold is breached. Second, the information field $I(s)$ is parameterized phenomenologically; future work should link this directly to single-cell transcriptomic atlases of the developing spine. Third, the protein dynamics model simplifies the complex metabolic network into a two-component system (demand vs. supply); while this captures essential scaling behavior, it abstracts away specific pathway kinetics. We emphasize that the demand–supply structural anisotropy gap is highly exploratory and is observed only in a narrow, carefully curated subset of extended-filament/channel mechanosensors ($n = 7$) versus metabolic regulators ($n = 7$), rather than representing a general property of all mechanotransduction-tagged proteins (which dilutes the signal). Fourth, the cross-species $\mathcal{B}_g$ analysis relies partly on estimated stiffness values for species where direct measurements are unavailable; experimental verification across additional species would strengthen the universality claim. Finally, the AlphaFold v6 structure for PIEZO2 covers only residues 1–709 of the full 2822-residue protein, and full-length structural analysis may modify the anisotropy estimate. A key limitation of the AlphaFold structural mapping lies in protein confidence scoring. Seven out of the 27 modeled proteins (including critical targets such as POC5, GHR, and MESP2) displayed lower confidence scores (pLDDT < 70). These lower scores predominantly reflect Intrinsically Disordered Regions (IDRs). While low confidence often complicates structural rigidity metrics, it provides mechanistic insight: highly disordered proteins naturally map to the Γ_m metabolic supply cohort. Such intrinsic disorder is a classical functional signature of multi-functional signaling hubs, acting as adaptive metabolic rheostats rather than stiff, load-bearing η_p mechanosensors. Expanding the supply-side AFCC panel beyond $n = 7$ is the highest-priority structural analysis next step. The current clinical evidence in this repository is a synthetic calibration against published trial rates plus an open SpineWeb/AASCE landmark-geometry check; it does not include retrospective or prospective patient records. Section 5.5 outlines the study design required for definitive clinical validation.

6 Conclusions

We have identified the Allometric Trap — a universal scaling constraint governing the metabolic cost of maintaining body shape against gravity — and demonstrated, in this model, that adolescent idiopathic scoliosis is a predictable consequence of crossing this boundary during growth. The Bio-Gravitational Number $\mathcal{B}_g = EI/MgL^2$ provides a simple criterion for the passive-to-active transition in upright postural maintenance, applied across 12 vertebrate species spanning four orders of magnitude in body mass (extension to bipedal/facultatively-bipedal species beyond hominins is an explicit next step). Exploratory AlphaFold structural analysis generates hypotheses regarding the molecular architecture of this vulnerability, suggesting a potential demand-supply structural gap between mechanosensory and metabolic proteins. The Delay Differential Equation (DDE) control framework unifies mathematical control theory, biomechanics, and developmental metabolism under a single structure, generating three falsifiable predictions testable with existing technology. These results motivate a research agenda in which the metabolic supply available during the adolescent growth spurt is examined as a candidate modifiable risk factor for AIS progression. We do not propose, on the basis of in-silico modeling alone, any specific clinical intervention; the framework’s value at this stage is in identifying candidate mechanisms and generating falsifiable predictions that prospective IRB-approved trials can test.

Code and Data Availability

All simulations and analyses were performed using the open-source Python package `spinalmodes` (v0.4.0), available at <https://github.com/sayujks0071/scoliosis> (current submission tag: `v1.4-editorial`; the full revision history including the initial submission snapshot `v1.0-submission` and the fragility-transparency revision `v1.3-fragility-transparent` is preserved as git tags). The Zenodo archival DOI will be minted from the GitHub release after acceptance (metadata in `.zenodo.json`). The computational environment, including dependencies such as PyElastica, JAX, NumPy, and optional `warp-lang` for GPU screening experiments, is specified in `pyproject.toml` and `requirements.txt`. All figure data are stored as CSV files and can be regenerated from the wrapper scripts in `scripts/experiments/`. The NVIDIA Warp screening outputs are stored under `results/nvidia_warp_beam_sweep/` and are reproducible from `scripts/experiments/experiment_nvidia_warp_beam_sweep.py`. AlphaFold structural metrics were computed from AlphaFold Protein Structure Database snapshots cached in `research/alphafold_counterscurvature/data/` (manifest in `manifest.csv`); the analysis is reproducible from those cached PDBs. **No human-subject data, patient records, or proprietary clinical datasets were used. All correlations reported in this manuscript are model-internal** (simulation outputs analyzed against simulation parameters or against publicly available comparative-anatomy data); prospective external validation against AIS clinical cohorts is identified as future work.

Ethics Approval

This computational study involves no human subjects, animal experiments, or clinical patient data. All analyses are based on publicly available allometric scaling relationships and published structural parameters (AlphaFold Protein Structure Database, comparative-anatomy datasets). No ethical approval from an Institutional Review Board or animal-care committee was required.

Consent to Participate

Not applicable. This study involved no human participants.

Consent to Publish

Not applicable. This study contains no individually identifiable data, images, or case material requiring consent for publication.

Competing Interests

The author declares no competing financial or non-financial interests.

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Author Contributions

S.K. conceived the theoretical framework, designed and performed all computational analyses, developed the software implementation, interpreted the results, and wrote the manuscript.

Tables

Table 3: Cross-species Bio-Gravitational Number $\mathcal{B}_g = EI/(MgL^2)$. Species with $\mathcal{B}_g < 0.1$ (shaded) require active metabolic maintenance of spinal geometry. Humans occupy the deepest position in the Allometric Trap.

Species	Mass (kg)	Length (m)	EI (Nm ²)	$\mathcal{B}_g$	Posture
Mouse	0.03	0.05	0.00008	10.9	Quadruped
Rat	0.3	0.15	0.002	3.02	Quadruped
Rabbit	2.5	0.35	0.03	0.99	Quadruped
Cat	4.0	0.45	0.06	0.76	Quadruped
Dog	20.0	0.75	0.5	0.45	Quadruped
Chimpanzee	50.0	0.5	1.6	0.65	Facultative biped
Human (child)	30.0	0.45	1.2	0.06	Biped
Human (adult)	70.0	0.6	2.5	0.01	Biped
Horse	500.0	1.5	40.0	0.36	Quadruped
Elephant	4000.0	2.5	550.0	0.22	Quadruped

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Table 4: Computational model parameters and biological justification

Parameter	Value	Units	Biological Basis
<i>Geometry</i>			
Length L	0.40	m	Adult spine (sacrum to cranium)
Diameter d	0.03	m	Typical vertebral body dimension
n elements	50–100	–	Convergence-tested discretization
<i>Material Properties</i>			
Young’s modulus E_0	1.0	GPa	Effective modulus (bone + disc)
Area moment I_{area}	1.0×10^{-8}	m^4	Cylindrical cross-section
Density ρ	1100	kg/m^3	Tissue-averaged density
Cross-section A	7.1×10^{-4}	m^2	$\pi(d/2)^2$
<i>IEC Coupling Parameters</i>			
χ_κ	0–0.10	m^{-1}	Rest curvature coupling (sweep)
χ_E	0.10	–	Stiffness modulation (fixed)
χ_M	0–0.05	N·m	Active moment coupling
β_1, β_2	0.1, 0.05	–	Metric coupling constants
<i>Information Field (Spinal Profile)</i>			
Cervical peak	$s/L = 0.80$	–	C1-C7 region
Lumbar peak	$s/L = 0.25$	–	L1-L5 region
Peak amplitude	0.5–0.7	–	Normalized HOX expression
Baseline I_0	0.3	–	Background patterning
<i>Loading & Dynamics</i>			
Gravity g	0.01–1.0	$g_\oplus$	Microgravity to Earth
Damping ν	0.1–1.0	s^{-1}	Numerical dissipation
Equilibrium criterion	$v_{\text{max}} < 10^{-6}$	m/s	Kinetic energy threshold

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Table 5: Thermodynamic cost proteins ($n = 23$) classified as Demand (mechanosensory) or Supply (metabolic). Anisotropy values from AlphaFold Database v6. The exploratory gap between Demand and Supply anisotropy ($p = 0.011$, Mann–Whitney U; Cohen’s $d = 1.19$) suggests a hypothesized molecular basis of the Energy Deficit Window. *PIEZO2 v6 structure covers residues 1–709 of 2822; full-length anisotropy may differ.

Gene	UniProt	Category	Anisotropy	Disorder %	pLDDT	Residues
<i>Demand-Side: Mechanosensory and Cytoskeletal (mean anisotropy 3.08 ± 1.44)</i>						
VIM	P08670	Demand	5.57	10.1	77.1	466
PTK7	Q13308	Demand	5.28	15.2	74.3	1070
LMNA	P02545	Demand	4.71	12.4	76.4	664
DSTYK	Q6XUX3	Demand	3.65	18.7	70.2	929
PIEZO2*	Q9H5I5	Demand	3.45	8.3	79.4	709
PIEZO1	Q92508	Demand	3.14	11.5	72.0	2521
CAV1	Q03135	Demand	2.52	22.4	78.4	178
FLNA	P21333	Demand	2.50	9.8	76.5	2647
RUNX3	Q13761	Demand	2.06	35.2	60.6	415
NTRK3	Q16288	Demand	1.94	14.1	76.8	839
EGR3	Q06889	Demand	1.56	64.1	50.0	387
LBX1	P52954	Demand	1.36	42.3	66.9	281
<i>Supply-Side: Metabolic Regulators (mean anisotropy 1.79 ± 0.56)</i>						
COL1A1	P02452	Supply	2.80	48.6	52.7	1464
SHH	Q15465	Supply	2.12	18.4	78.4	462
SOX9	P48436	Supply	2.19	44.0	56.0	509
CDKN1A	P38936	Supply	2.14	38.9	69.0	164
COMP	P49747	Supply	1.72	6.2	88.1	757
SIRT1	Q96EB6	Supply	1.73	35.1	65.0	747
PPARGC1A	Q9UBK2	Supply	1.52	61.9	52.7	798
ARNTL	O00327	Supply	1.48	42.7	58.3	626
ACAN	P16112	Supply	1.35	28.4	71.2	2415
MMP3	P08254	Supply	1.28	12.1	82.4	477
MMP1	P03956	Supply	1.22	10.8	84.1	469

Table 6: The 16 candidate proteins analyzed in this study, grouped by mechanistic role (η_p proprioceptive channels, η_a active control, Γ_m metabolic regulators) and characterized by AlphaFold-derived structural metrics. The anisotropy ratio and morphology classification indicate the relative structural cost of maintaining directional sensing versus volumetric (scalar) function. Genes are prioritized by confidence-weighted scoring from the AFCC pipeline.

Gene	UniProt	Category	Anisotropy	Morphology	pLDDT	Scaling
<i>Proprioceptive Channel Anchors (η_p)</i>						
PIEZO2	Q9H5I5	η_p	4.44	Fibrous/Extended	79.4	L
PIEZO1	Q92508	η_p	3.90	Fibrous/Extended	72.0	L^2
EGR3	Q06889	η_p	3.76	Fibrous/Extended	50.0	L
RUNX3	Q13761	η_p	2.06	Intermediate	60.6	L
<i>Active Maintenance Anchors (η_a)</i>						
LMNA	P02545	η_a	4.75	Fibrous/Extended	76.4	L^2
FLNA	P21333	η_a	2.50	Intermediate	76.5	L^3
LBX1	P52954	η_a	2.27	Intermediate	66.9	L^2
MYLK	Q15746	η_a	1.46	Globular	65.8	L^2
DMD	P11532	η_a	1.32	Globular	76.3	L^3
VIM	P08670	η_a	7.47	Fibrous/Extended	77.1	L^3
<i>Metabolic Supply Scaling (Γ_m)</i>						
GHR	P10912	Γ_m	5.13	Fibrous/Extended	58.7	L
ARNTL	O00327	Γ_m	3.32	Fibrous/Extended	65.5	L
COL1A1	P02452	Γ_m	2.80	Intermediate	52.7	L^3
PPARGC1A	Q9UBK2	Γ_m	2.19	Intermediate	52.7	L
SOX9	P48436	Γ_m	2.19	Intermediate	56.0	L
IGF1R	P08069	Γ_m	1.43	Globular	78.0	L

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Table 7: Comprehensive Statistical Summary of the Information-Elasticity Coupling (IEC) Framework. All statistical tests were independently computed using *scipy.stats* to summarize model-internal computational claims.

Mechanism / Hypothesis	Metric Evaluated	Statistical Result	Significance Level
<i>Cross-Species Scaling & The Allometric Trap</i>			
Allometric Scaling	B _g vs. Body Mass	Exponent = -0.282 ± 0.072 $R^2 = 0.744$ Deviation from -0.25 : 0.032	$p = 1.31 \times 10^{-3}$ (Strong)
<i>Energy Deficit Window & Scoliosis Onset</i>			
Critical Transition (L_{crit})	Spine Length vs. simulated Cobb Angle	$r = 0.983$ Deficit Ratio Max = 38.5	$p = 2.74 \times 10^{-22}$ (Model-internal)
Anisotropy Rescue Effect	L_{crit} vs. Anisotropy	$R^2 = 0.775$ Saturation at ≥ 6.0	$p = 3.58 \times 10^{-17}$ (Model-internal)
<i>Mechanical Instability & Directional-Stiffness Anisotropy</i>			
Directional Misalignment	Peak Stiffness Mismatch	Position = 0.596 (Normalized) Reduction = 31.1%	Matching Thoracolumbar Apex (Verified)

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Supplementary Information

S1 Cosserat Rod Geometry and Kinematic Parameterization

We model the human spine as a Cosserat rod, defined by a centerline curve $\mathbf{r}(s, t) \in \mathbb{R}^3$ parameterized by arc length $s \in [0, L]$. This dimensionality reduction from 3D elasticity to a 1D director theory is justified by the high aspect ratio of the vertebral column ($L/d \approx 13$). To each point s , we attach an orthonormal material frame $\{\mathbf{d}_1(s, t), \mathbf{d}_2(s, t), \mathbf{d}_3(s, t)\}$ describing the cross-sectional orientation [8, 32]. The kinematics are governed by:

$$\mathbf{r}'(s) = \mathbf{v}(s) = \mathbf{d}_3(s) + \boldsymbol{\varepsilon}(s), \quad (\text{S1})$$

$$\mathbf{d}_i'(s) = \mathbf{u}(s) \times \mathbf{d}_i(s), \quad (\text{S2})$$

where $\mathbf{v}(s)$ is the tangent vector, $\boldsymbol{\varepsilon}(s)$ represents shear and extension strains, and $\mathbf{u}(s) = \kappa_1 \mathbf{d}_1 + \kappa_2 \mathbf{d}_2 + \tau \mathbf{d}_3$ is the Darboux vector encoding flexural curvature $\kappa_{1,2}$ and torsional twist τ .

Boundary Conditions. The sacral base ($s = 0$) is clamped and the cranial tip ($s = L$) is free:

$$\begin{aligned} \mathbf{r}(0) &= \mathbf{0}, \quad \mathbf{d}_i(0) = \mathbf{e}_i \quad (\text{Clamped Base}) \\ \mathbf{n}(L) &= \mathbf{0}, \quad \mathbf{m}(L) = \mathbf{0} \quad (\text{Free Tip}) \end{aligned} \quad (\text{S3})$$

S2 Cosserat Force and Moment Balance

For a static rod subject to gravity $\mathbf{f}_g = \rho A \mathbf{g}$ and active moments induced by the information field, the balance equations for internal force $\mathbf{n}$ and moment $\mathbf{m}$ are:

$$\begin{aligned} \mathbf{n}'(s) + \mathbf{f}_g &= \mathbf{0}, \\ \mathbf{m}'(s) + \mathbf{r}'(s) \times \mathbf{n}(s) + \mathbf{m}'_{\text{info}}(s) &= \mathbf{0}, \end{aligned} \quad (\text{S4})$$

where $\mathbf{m}_{\text{info}}$ represents the active couple generated by the information field gradient.

S3 Developmental Information Field Parameterization

The scalar information field $I(s)$ is parameterized as a bimodal Gaussian:

$$I(s) = A_c \exp \left[-\frac{(s/L - s_c)^2}{2\sigma_c^2} \right] + A_l \exp \left[-\frac{(s/L - s_l)^2}{2\sigma_l^2} \right] + I_0, \quad (\text{S5})$$

where $A_c = 0.5$, $s_c = 0.80$, $\sigma_c = 0.08$ (cervical lordosis), $A_l = 0.7$, $s_l = 0.25$, $\sigma_l = 0.10$ (lumbar lordosis), and $I_0 = 0.3$ (baseline).

The information content is quantified using the local Shannon entropy:

$$\mathcal{H}[I] = - \int I(s) \log I(s) ds, \quad (\text{S6})$$

which bounds the precision of the target metric specification.

S4 Biological Metric and Information Geometry

The central hypothesis of the IEC framework is that developmental information modifies the effective metric experienced by the rod's reference manifold. We define a biological metric:

$$d\ell_{\text{eff}}^2 = g_{\text{eff}}(s) ds^2 = \exp \left[2 \left(\beta_1 \tilde{I}(s) + \beta_2 \frac{\partial \tilde{I}}{\partial s} \right) \right] ds^2, \quad (\text{S7})$$

where $\tilde{I}$ is the normalized information field and $\beta_{1,2}$ are dimensionless coupling constants.

S5 Equilibrium Deviation and Regime Classification

We quantify the degree of biological counter-curvature using the Kullback–Leibler divergence between the IEC-coupled geometry and the passive gravitational state:

$$D_{\text{KL}} = \int_0^L \left[\frac{1}{2} (\kappa_{\text{IEC}}(s) - \kappa_{\text{passive}}(s))^2 \mathcal{F}(s) \right] w(s) ds, \quad (\text{S8})$$

where $\mathcal{F}(s)$ is the local Fisher Information and $w(s) = g_{\text{eff}}(s)$. The normalized deviation from the gravity-only equilibrium is:

$$\hat{D}_{\text{geo}} = \frac{D_{\text{KL}}}{D_{\text{KL}}^{\text{max}}(\chi_{\kappa}, g)}. \quad (\text{S9})$$

Regime thresholds: gravity-dominated ($\hat{D}_{\text{geo}} < 0.1$), cooperative ($0.1 \leq \hat{D}_{\text{geo}} \leq 0.3$), information-dominated ($\hat{D}_{\text{geo}} > 0.3$). These were determined by analyzing the bifurcation of the lowest eigenvalue λ_0 .

S6 Metabolic Power Density and Dissipation

The metabolic power density required to sustain counter-curvature is:

$$\dot{\mathcal{F}} = \int_0^L \left[\eta_p \left| \frac{\partial \kappa}{\partial t} \right|^2 + \eta_a (\kappa(s) - \kappa_{\text{passive}}(s))^2 + \Gamma_m(s) \right] ds, \quad (\text{S10})$$

where η_p is the proprioceptive feedback dissipation coefficient, η_a is the active moment maintenance coefficient, and $\Gamma_m(s)$ is the basal tissue maintenance cost density.

S7 Protein Dynamics and Energy Competition

During adolescent growth, mechanosensory and growth/maintenance protein classes compete for a finite metabolic energy pool:

$$E_{\text{total}}(t) = E_{\text{mechano}}(t) + E_{\text{growth}}(t) + E_{\text{metabolic}} \quad (\text{S11})$$

$$\frac{dE_{\text{mechano}}}{dt} = k_{\text{syn},m} \cdot [\text{Resources}] - k_{\text{deg},m} \cdot [\text{Mechano}] - \dot{\mathcal{F}}_{\text{load}} \quad (\text{S12})$$

where $\dot{\mathcal{F}}_{\text{load}} \propto L^4$ represents the increasing thermodynamic load. The growth protein synthesis rate is driven by GH/IGF-1 and scales with L^2 (surface area of nutrient diffusion), creating the scaling mismatch at the molecular level.

The vulnerability is quantified via:

$$\Lambda(t) = \frac{dV/dt}{\tau_{\text{adapt}}} \quad (\text{S13})$$

where dV/dt is volumetric growth rate and τ_{adapt} is the mechanosensory adaptation time constant.

S8 The Spinal Clock and Convective Shutdown

The intervertebral disc possesses an autonomous circadian clock driven by the BMAL1/CLOCK transcriptional loop [46]. Gravity provides the synchronization signal through daily loading cycles: daytime compression forces fluid out of the nucleus pulposus, while nocturnal unloading draws fluid back via osmotic pressure [27].

Under microgravity or prolonged bed rest, this pump arrests (“Convective Shutdown”), creating hypoxia that stabilizes HIF-1 α and upregulates catabolic enzymes (MMP1, MMP3). The resulting matrix degradation reduces $B_{\text{eff}}(s)$ and widens the Energy Deficit Window.

This mechanism generates three falsifiable predictions:

1. Clock gene desynchronization (Per2, Bmal1 flattening in IVD) within 72 hours of unloading, preceding structural changes.
2. HIF-1 α stabilization in the nucleus pulposus preceding scoliotic curvature; HIF-1 α inhibitors should delay onset.
3. Pulsatile axial compression at the appropriate circadian phase will preserve spinal geometry more effectively than static compression.

S8.1 Exploratory Spinal-Jetlag Simulation

As an exploratory extension, we incorporated circadian variation into the information-coupling term and compared an entrained baseline with a high- χ_κ phase-shifted condition. The phase-shifted simulation produced a 2.52-fold increase in mean simulated Cobb angle compared with the entrained baseline (24.18° vs. 9.61°; Mann–Whitney U test, $p = 2.59 \times 10^{-4}$). This result is retained as a supplementary hypothesis-generating analysis only. It is not used as a core claim of the main-text model and requires independent validation against disc-clock, actigraphy, and longitudinal AIS imaging data before clinical interpretation.

S9 GPU Screening and PyElastica Refinement Workflow

We implemented an optional NVIDIA Warp workflow to accelerate broad passive-mechanics parameter screening before selecting cases for nonlinear Cosserat-rod refinement. The screening kernel evaluates the Euler–Bernoulli cantilever tip-sag approximation, the Bio-Gravitational Number $\mathcal{B}_g$, the passive sag ratio, and a dimensionless triage score across deterministic combinations of spine length (0.25–0.55 m), radius (0.007–0.014 m), effective Young’s modulus (10^6 – 1.5×10^9 Pa), and tissue density (900–1200 kg/m³). This workflow is computational only: it is designed to identify parameter regimes requiring full nonlinear simulation, not to generate clinical predictions.

The command

```
python scripts/experiments/experiment_nvidia_warp_beam_sweep.py \
  --n-samples 100000 \
  --device auto \
  --out-dir results/nvidia_warp_beam_sweep \
  --max-csv-rows 10000 \
  --refine-pyelastica \
  --refine-count 3 \
  --refine-elements 20 \
  --refine-final-time 0.5 \
  --refine-max-sag-ratio 2.0
```

was run on an NVIDIA GB10 GPU using `warp-lang` 1.13.0. The Warp kernel evaluated 100,000 candidate parameter sets in 0.0057s after compilation (approximately 1.74×10^7 samples/s). The screened distribution had median $\mathcal{B}_g = 27.41$, minimum $\mathcal{B}_g = 0.0083$, and 0.127% of candidates below the $\mathcal{B}_g < 0.1$ low-passive-stability threshold. The 95th percentile passive sag ratio was 0.049.

The bounded refinement stage selected high-risk rows with analytic sag ratio ≤ 2.0 and reran three cases through the validated PyElastica beam setup. All three PyElastica refinements

completed successfully. The refined cases had $\mathcal{B}_g \approx 0.063$ and analytic sag ratios near 2.0; their large discrepancy from the small-deflection analytic formula (maximum relative error 0.991) confirms that the Warp screen correctly identifies regimes where the analytic small-deflection approximation is no longer sufficient and nonlinear Cosserat refinement is required. These GPU-screening results are therefore reported as a reproducibility and triage tool rather than as a biological validation result.

S10 Complete Theoretical Derivations

This section provides the complete mathematical derivations, delay differential equation analysis, Hopf bifurcation calculations, and detailed coupling mechanism formulations supporting the main-text conceptual summary in Section 2.

S10.1 The Bio-Gravitational Number and the Allometric Trap

For a vertebral column of length L , flexural stiffness EI , mass M , and gravitational acceleration g , we define the **Bio-Gravitational Number**:

$$\mathcal{B}_g = \frac{EI}{MgL^2}, \quad (\text{S14})$$

a dimensionless ratio of passive flexural resistance to gravitational bending moment. When $\mathcal{B}_g > 0.1$, an organism can resist gravitational collapse through passive stiffness alone. When $\mathcal{B}_g < 0.1$, the organism enters the ‘‘Allometric Trap’’: it must actively maintain its shape at continuous metabolic cost. Cross-species analysis reveals that small quadrupeds operate well above this threshold, while humans ($\mathcal{B}_g \approx 0.01$) are deep within the trap, relying almost entirely on active metabolic maintenance to prevent spinal collapse [47]. The Bio-Gravitational Number is analogous to the Froude number for locomotion, defining a fundamental physical boundary for spinal stability. However, scaling exponents vary systematically depending on physiological state [48], and vertebral scaling exhibits size-dependent breakpoints [49], emphasizing that bipedalism and human-specific axial loading uniquely exacerbate this vulnerability [50].

The central problem of vertebrate morphogenesis is the translation of a one-dimensional genetic code into a robust three-dimensional geometry that can withstand environmental loads. Classical structural mechanics predicts that a flexible column clamped at the base and subjected to its own weight will buckle or sag into a monotonic C-shape [8]. Yet, the healthy human spine exhibits a complex, alternating S-shape (lordosis and kyphosis).

This anomaly indicates that the S-shape is not a passive equilibrium but is actively maintained by continuous metabolic expenditure against the gravitational field. This maintenance is naturally framed as a biological instance of *optimal feedback control* [51]: the nervous system minimizes a cost function combining geometric error (deviation from the target shape) and metabolic effort (ATP consumption).

S10.2 Delay Differential Equations (DDE) and the Derivative Gain Trap

The active maintenance of spinal alignment relies on proprioceptive feedback from paraspinal muscles and mechanosensors. However, neural conduction and central processing are not instantaneous. Established models of human postural control demonstrate that the upright state is stabilized by proportional-derivative (PD) or proportional-derivative-acceleration (PDA) feedback, subject to a time delay τ [23, 24]. We formalize the spine as an active control system governed by a Delay Differential Equation (DDE).

The proprioceptive control law generating corrective force F_{control} with feedback delay τ is:

$$\kappa_{\text{rest}}(s) = \kappa_{\text{gen}} + \chi_{\kappa} \nabla I(s). \quad (\text{S15})$$

where K_P and K_D are proportional and derivative gains, and the error is $e(s, t) = \kappa_{measured}(s, t) - \kappa_{target}$.

Linearizing around the straight configuration, the dynamics of the spinal column become:

$$\rho A \frac{\partial^2 \kappa}{\partial t^2} + \eta \frac{\partial \kappa}{\partial t} + EI \frac{\partial^4 \kappa}{\partial s^4} + K_P \kappa(s, t - \tau) + K_D \frac{\partial \kappa}{\partial t}(s, t - \tau) = 0 \quad (S16)$$

The stability of this active structure is governed by the dimensionless Bio-Gravitational Number $\mathcal{B}_g$:

$$\mathcal{B}_g = \frac{\chi_M \langle |\nabla I| \rangle}{\rho A g L^2}. \quad (S17)$$

Crucially, when $\mathcal{B}_g > 1$, the biological system effectively dominates gravitational mechanics. At the molecular level, this coupling relies on the structural rigidity of patterning transcription factors; AlphaFold structural analysis of critical HOX proteins (e.g., HOXC8, UniProt P31273; HOXB13, UniProt Q92826) reveals conserved, high-confidence DNA-binding domains capable of enforcing the sharp spatial identity boundaries required by our field parameterization.

During the adolescent growth spurt, the spine elongates rapidly. Because nerve conduction velocity in the extremities does not scale proportionately with rapid height increases [25, 52], the absolute neural feedback delay τ increases significantly. We term this vulnerability the **Derivative Gain Trap**: derivative gains (K_D) that provide essential damping and stability during childhood become destabilizing as the delay τ crosses a critical threshold τ_c .

S10.3 Hopf Bifurcation and the Onset of Scoliosis

When the neural delay exceeds the critical threshold ($\tau > \tau_c$), the system undergoes a supercritical Hopf bifurcation [26]. The previously stable upright equilibrium becomes unstable, and stable limit cycles (oscillations) emerge. In the context of the spine, these delay-induced oscillations manifest as continuous, micro-mechanical buckling events that asymmetric biomechanical loading then locks into a permanent scoliotic deformity.

Assuming solutions of the form $\kappa(s, t) = A e^{\lambda t} \sin(\frac{n\pi s}{L})$, the characteristic equation is:

$$\kappa_{rest}(s) = \kappa_{gen} + \chi_\kappa \nabla I(s), \quad (S18)$$

where χ_κ is a geometric coupling constant and κ_{gen} is the baseline genetic curvature.

Stiffness modulation (IEC-2). The information field modulates the effective bending and torsional stiffness:

$$B_{eff}(s) = E_0 I_{area} (1 + \chi_E I(s)), \quad C_{eff}(s) = G_0 J (1 + \chi_C I(s)). \quad (S19)$$

The total energy functional governing the rod's equilibrium is:

$$\mathcal{E}_{total} = \int_0^L \left[\frac{1}{2} B_{eff} (\kappa - \kappa_{rest})^2 + \frac{1}{2} C_{eff} \tau^2 + \frac{1}{2} K_{eff} \epsilon^2 + \rho A \mathbf{r} \cdot \mathbf{g} \right] ds. \quad (S20)$$

For purely imaginary roots $\lambda = \pm i\omega$ (defining the bifurcation boundary), the system transitions from damped stability to active oscillation. This mathematical control theory precisely explains why Adolescent Idiopathic Scoliosis (AIS) uniquely coincides with the rapid adolescent growth spurt: it is the exact developmental window where increasing τ pushes the neuromotor control system across the Hopf bifurcation boundary.

S10.4 The Energy Deficit Window: NAD+ and Proprioceptive Integrity

The biological spine is a dissipative structure requiring continuous metabolic expenditure to maintain its sensory and structural integrity. The metabolic power required to sustain this configuration scales superlinearly with length (L^4 demand vs L^2 surface-limited supply) [28].

This scaling mismatch creates a transient **Energy Deficit Window** during the adolescent growth spurt. We propose that this metabolic deficit directly modulates the control parameters in the DDE model. Specifically, proprioceptive Ia/II afferents are among the largest and most metabolically demanding neurons.

Recent evidence demonstrates that NAD+ depletion during development causes severe spinal deformities [29], while oxidative stress directly drives scoliosis in animal models [53]. Furthermore, NAD+ insufficiency activates the SARM1 pathway, which triggers local axon degeneration [30]. We hypothesize that the adolescent Energy Deficit Window causes localized NAD+ insufficiency, which mildly compromises proprioceptive axon integrity. This subclinical neuropathy further increases the effective neural delay τ , accelerating the transition across the Hopf bifurcation boundary and triggering scoliotic buckling.

S10.5 Molecular Grounding: The AlphaFold Anisotropy Gap (Exploratory)

While the primary failure mode is governed by macroscopic delay dynamics, we also explore the molecular architecture of the mechanosensory apparatus. AlphaFold v6 structural analysis of 23 key spinal proteins reveals that demand-side mechanosensory proteins (e.g., Vimentin, PIEZO2) exhibit significantly higher structural anisotropy (mean 3.08 ± 1.44) than supply-side metabolic regulators (mean 1.79 ± 0.56).

We tentatively propose a **Demand-Supply Anisotropy Gap** ($p = 0.011$ in our dataset): the proteins required to *sense* mechanical geometry are structurally extended and thermodynamically expensive to maintain, while their metabolic regulators are compact and often intrinsically disordered. We emphasize that this observation is hypothesis-generating. Current AI structural prediction tools cannot directly infer mechanical properties under load [35, 54], and confidence scores for extracellular matrix components are frequently low. Nonetheless, this structural asymmetry provides a compelling direction for future experimental validation of the cellular mechanisms underlying the Energy Deficit Window.

S11 Detailed Methods

This section provides detailed computational protocols, steered molecular dynamics parameters, tissue homogenization calculations, and protein dynamics modeling supporting the main-text Methods summary in Section 3.

S11.1 Multi-Scale Protein Mechanics Pipeline

To ground the macroscopic parameters η_p, η_a, Γ_m in molecular reality, we developed a multi-scale pipeline integrating AlphaFold predictions with structural energetics.

1. **Structure Prediction:** We retrieved high-confidence structures for 23 key spinal proteins from the AlphaFold Protein Structure Database.
2. **Structural Metrics:** For each protein, we computed the *Anisotropy Index* (ratio of principal moments of inertia) and *Radius of Gyration* (R_g) using the `afcc` module. These geometric metrics are reported as exploratory structural descriptors of the AlphaFold-predicted static conformation, not as direct measurements of mechanical properties.

3. **Working hypothesis on metabolic cost:** As an exploratory hypothesis, we conjecture that the maintenance cost of a protein scales with $\text{Anisotropy} \times N_{\text{residues}}$ (extended structures plausibly require more chaperone intervention and proteostatic support). This is a working assumption, not a measured quantity, and is not load-bearing for the central control-theoretic argument; we use it only to motivate the demand-vs-supply categorization tested in Results.
4. **Steered Molecular Dynamics (planned, not in current submission):** A subsequent work in preparation will report SMD simulations (GROMACS, CHARMM36m, TIP3P) extracting force-extension curves and single-molecule stiffness k_{molecule} for the demand- and supply-side proteins, enabling direct comparison to published AFM nanoindentation data for PIEZO2 [10], vimentin, and lamin A/C. Quantitative validation of the anisotropy-cost mapping against single-molecule force spectroscopy is identified as the principal external validation step required before the molecular results in this paper can be elevated from exploratory to confirmatory.

S11.2 Protein Dynamics and Energy Modeling

We simulated the temporal competition between protein classes using a system of coupled differential equations. The model tracks the energy pool $E(t)$ available for protein synthesis/maintenance over the 5–20 year age range.

- **Supply:** Modeled as $S(t) = S_0(L(t)/L_0)^2$, reflecting the diffusion-limited transport through the vertebral endplate surface area.
- **Demand:** Modeled as $D(t) = D_{\text{maint}}(L/L_0) + D_{\text{grav}}(L/L_0)^4$, capturing the linear scaling of tissue volume maintenance and the quartic scaling of gravitational moment opposition.
- **Dynamics:** The fidelity of the mechanosensory array $F(t)$ evolves according to $dF/dt = k_{\text{repair}}(S - D)$ when $S > D$, and $dF/dt = k_{\text{damage}}(S - D)$ when $S < D$ (deficit), bounded to $[0, 1]$.

S11.3 Supplementary Table S2: Multiple-comparison correction across reported protein-statistic tests

Table S1: Benjamini–Hochberg false-discovery-rate adjusted q-values (and Bonferroni-corrected p-values, for reference) across the seven independent statistical tests reported in the protein-stratification analyses (Results §3.4). Tests ordered by raw p-value. Three tests survive the standard $q < 0.05$ FDR threshold (hinge density, anisotropy gap extended trimmed, anisotropy gap original); the biped/quadruped Mann–Whitney result is reported with the explicit caveat that it does not survive correction.

Test	N	p_{raw}	$p_{\text{Bonferroni}}$	q_{BH}
Mechanosensor hinge density (n=61)	61	0.0030	0.021	0.021
Anisotropy gap, extended cohort, trimmed	26	0.0094	0.066	0.026
Anisotropy gap, original demand/supply	23	0.0110	0.077	0.026
Biped vs Quadruped MWU (per-species)	12	0.0440	0.308	0.077
Disorder $\times$ anisotropy in mechanosensors	31	0.0720	0.504	0.101
AIS-strict-tagged vs other (text-match)	61	0.2340	1.000	0.273
AIS-strict-tagged trimmed	59	0.2870	1.000	0.287

Three additional robustness analyses (correlations #6, #7, #8 in the analysis pipeline at scripts/) provide further accountability:

- **Citation-bias check (correlation #6):** no correlation between curated clinical-evidence priority score (0–100) and AlphaFold anisotropy across $n = 61$ genes (Spearman $\rho = 0.016$, $p = 0.90$). The structural signal is mechanism-specific to the functional categorization, not a literature-popularity artifact.
- **Leave-one-out robustness (correlation #7):** for the demand-vs-supply MWU (baseline $p = 0.035$), 84.6% of LOO iterations remain $p < 0.05$ (LOO median $p = 0.041$, max $p = 0.092$). Result is borderline-fragile to single supply-side gene removal — reflects the small supply cohort ($n = 7$). For the hinge-density test (baseline $p = 0.0025$), 100% of LOO iterations remain $p < 0.01$ (LOO median $p = 0.003$, max $p = 0.0042$) — fully robust.
- **Demand-panel expansion sensitivity (correlation #8):** extending the demand panel from $n = 19$ curated extended-filament/channel proteins to all $n = 37$ mechanotransduction-tagged proteins dilutes the gap (relative-gap 93% $\rightarrow$ 42%, $p = 0.035 \rightarrow 0.33$). The published gap is therefore a property of extended-filament/channel mechanosensors specifically, not all mechanotransduction-tagged proteins. The headline numbers (72%, $p = 0.011$, Cohen’s $d = 1.19$) cited in Results §3.4 reference the curated $n = 23$ subset; the narrowed scope is reflected in the abstract and discussion.